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1 Section A

Executive summary

Global attempts to shift to cleaner and more efficient energy sources are thoroughly analyze the 'Energizing the Future' dataset, which provides a detailed investigation of sustainable energy trends from 2000 to 2020. The dataset covers access to energy, clean cooking fuels, renewable electricity generation, financial flows to developing nations, carbon emissions, and more with an emphasis on essential metrics. Insider knowledge about the worldwide low-carbon electrical landscape, per capita energy consumption, and the trajectory of renewable energy adoption is available to scholars, policymakers, and energy lovers. Understanding the developments, difficulties, and possible routes towards a sustainable energy future is made much easier with the help of this dataset. Notable attributes comprise past patterns, population figures, monetary commitments.

Key features include historical trends, per capita metrics, financial investments, and a holistic view of the renewable energy landscape, making it an essential tool for informed decision-making and strategic planning in the field of sustainable energy."

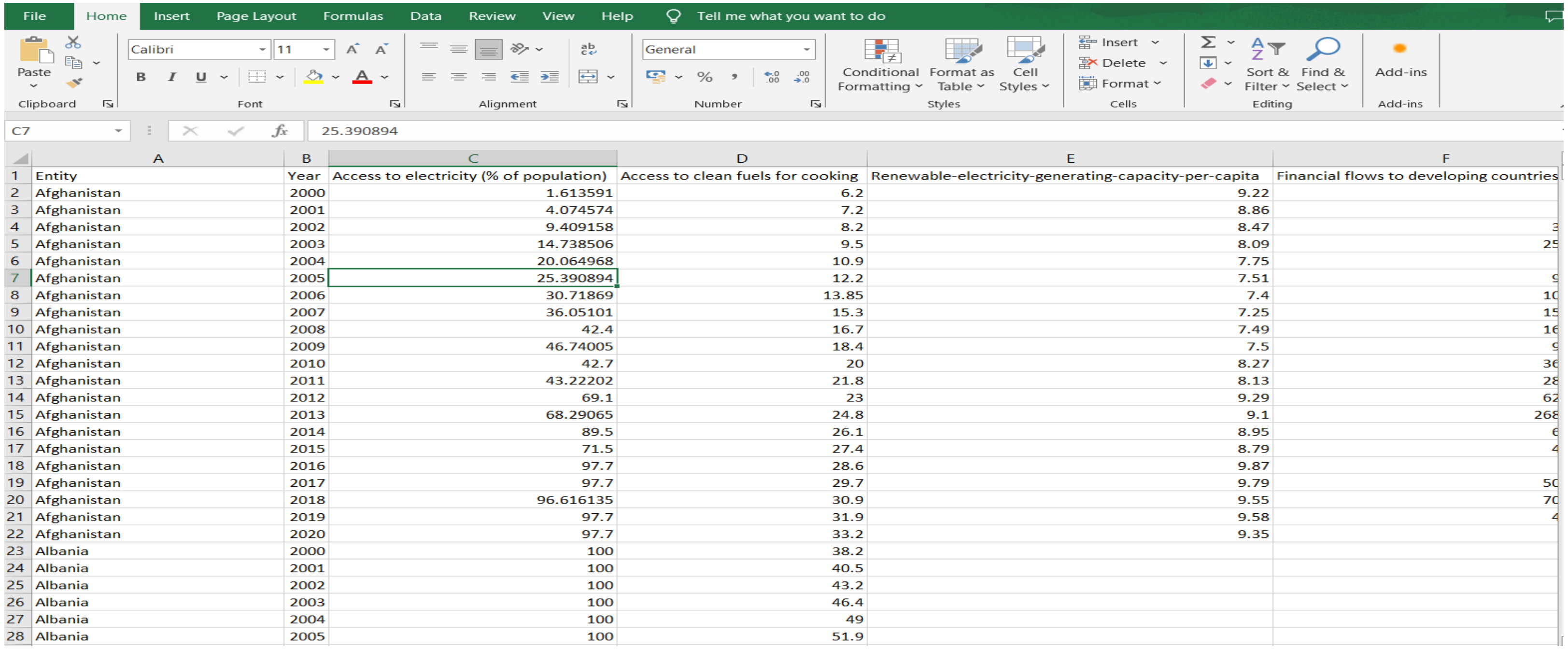
2 Section B

2.1 Introduction

A resilient and environmentally conscious future is within reach when we all pursue sustainable energy." A thorough summary of worldwide efforts in sustainable energy is given by the 'Energizing the Future' dataset, which covers the years 2000 to 2020. This dataset is a vital resource for scholars and decision-makers, focusing on important indicators like finances to developing countries, access to electricity, clean cooking fuels, and renewable energy capacities. In order to shape the course of our shared environmental legacy, our analysis will delve into the complexities of renewable energy trends, carbon emissions, and international investments in order to unearth insights that will guide strategic decisions and open the door to a more equitable and sustainable energy landscape.

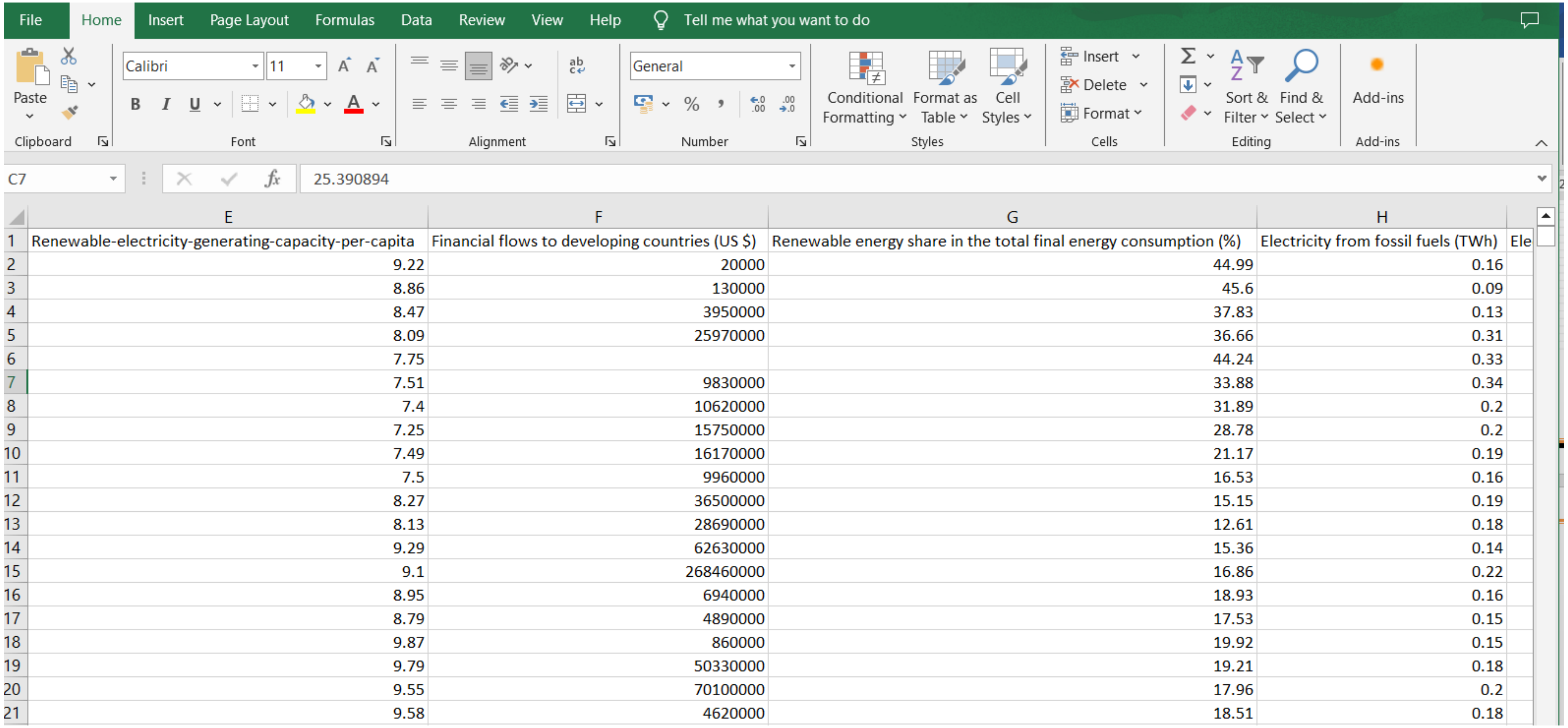
2.2 Dataset Description

This dataset provides a thorough record of sustainable energy trends from 2000 to 2020, offering a rich source of information for researchers and analysts. With 21 columns and 19 years of data, it covers crucial aspects such as access to electricity, clean cooking fuels, renewable energy capacities, financial flows, carbon emissions, GDP growth, and more. Researchers can explore patterns, correlations, and trends, gaining insights into the global landscape of sustainable energy. The dataset serves as a foundational tool for in-depth analysis, supporting investigations into the evolution and impact of sustainable energy practices on a global scale



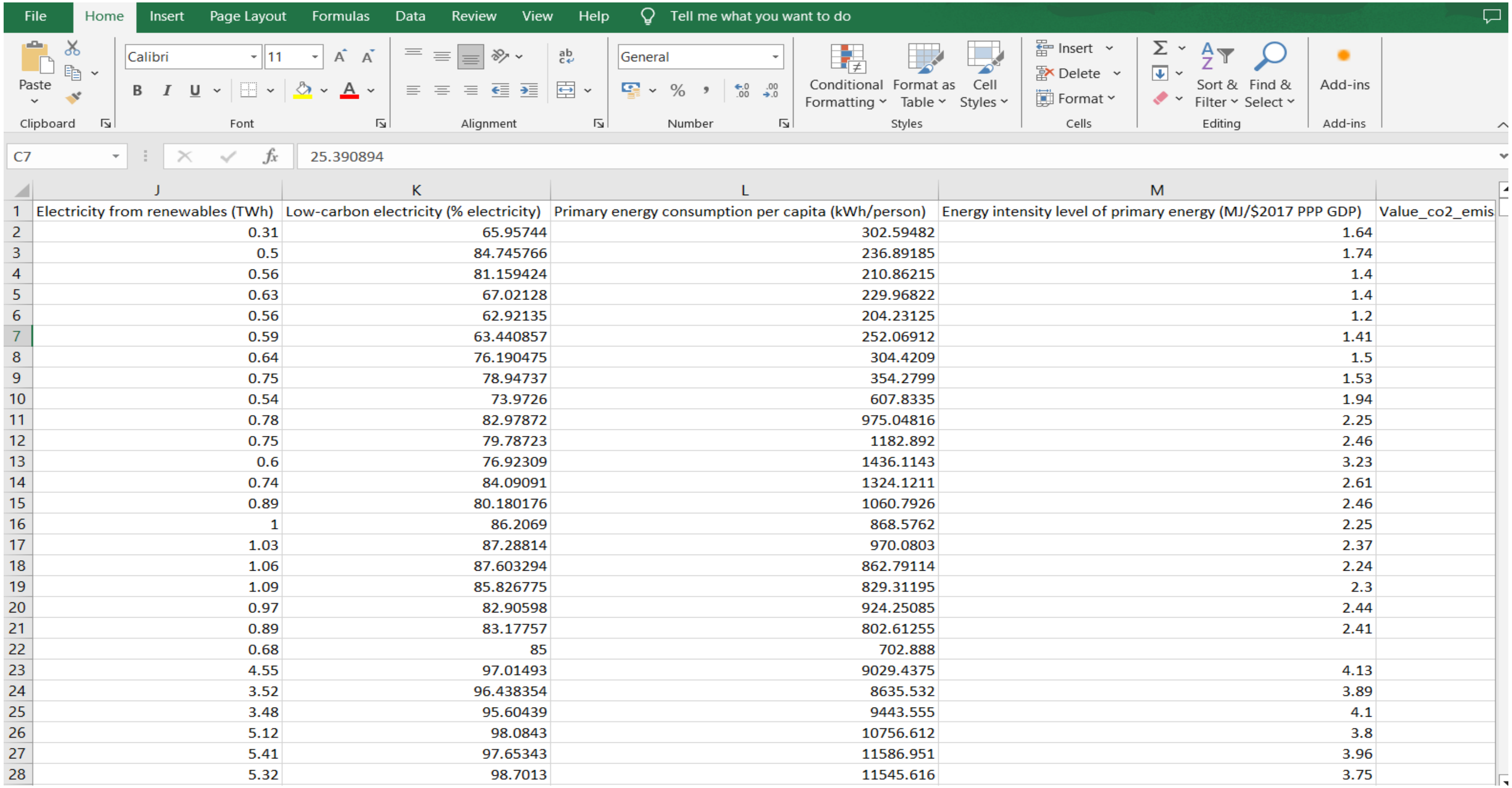
	A	B	C	D	E	F
1	Entity	Year	Access to electricity (% of population)	Access to clean fuels for cooking	Renewable-electricity-generating-capacity-per-capita	Financial flows to developing countries
2	Afghanistan	2000	1.613591	6.2	9.22	
3	Afghanistan	2001	4.074574	7.2	8.86	
4	Afghanistan	2002	9.409158	8.2	8.47	3
5	Afghanistan	2003	14.738506	9.5	8.09	25
6	Afghanistan	2004	20.064968	10.9	7.75	
7	Afghanistan	2005	25.390894	12.2	7.51	9
8	Afghanistan	2006	30.71869	13.85	7.4	10
9	Afghanistan	2007	36.05101	15.3	7.25	15
10	Afghanistan	2008	42.4	16.7	7.49	16
11	Afghanistan	2009	46.74005	18.4	7.5	9
12	Afghanistan	2010	42.7	20	8.27	36
13	Afghanistan	2011	43.22202	21.8	8.13	28
14	Afghanistan	2012	69.1	23	9.29	62
15	Afghanistan	2013	68.29065	24.8	9.1	268
16	Afghanistan	2014	89.5	26.1	8.95	6
17	Afghanistan	2015	71.5	27.4	8.79	4
18	Afghanistan	2016	97.7	28.6	9.87	
19	Afghanistan	2017	97.7	29.7	9.79	50
20	Afghanistan	2018	96.616135	30.9	9.55	70
21	Afghanistan	2019	97.7	31.9	9.58	4
22	Afghanistan	2020	97.7	33.2	9.35	
23	Albania	2000	100	38.2		
24	Albania	2001	100	40.5		
25	Albania	2002	100	43.2		
26	Albania	2003	100	46.4		
27	Albania	2004	100	49		
28	Albania	2005	100	51.9		

Figure 2.1 Dataset in excel sheet 1.1



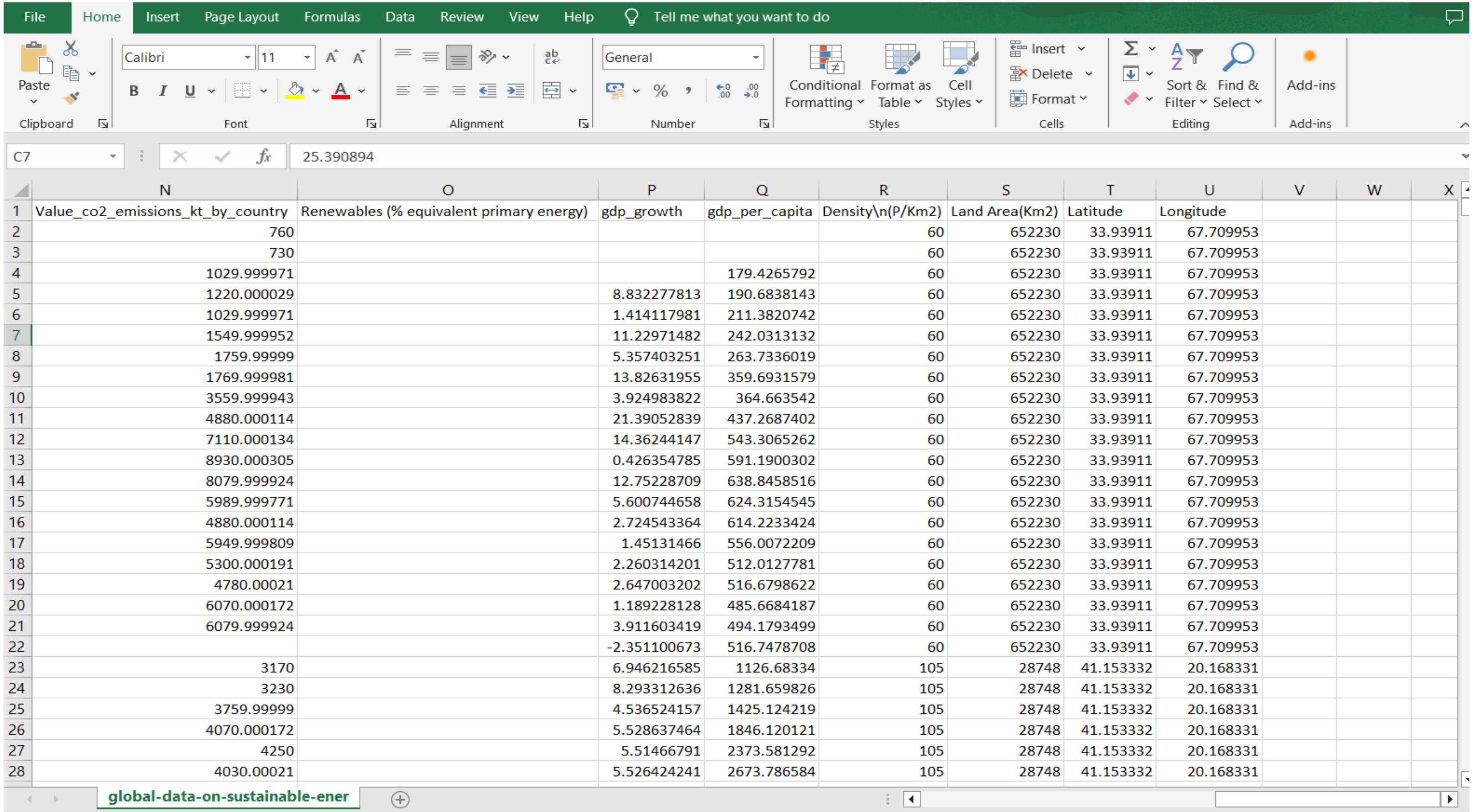
	E	F	G	H	
1	Renewable-electricity-generating-capacity-per-capita	Financial flows to developing countries (US $)	Renewable energy share in the total final energy consumption (%)	Electricity from fossil fuels (TWh)	Ele
2	9.22	20000	44.99	0.16	
3	8.86	130000	45.6	0.09	
4	8.47	3950000	37.83	0.13	
5	8.09	25970000	36.66	0.31	
6	7.75		44.24	0.33	
7	7.51	9830000	33.88	0.34	
8	7.4	10620000	31.89	0.2	
9	7.25	15750000	28.78	0.2	
10	7.49	16170000	21.17	0.19	
11	7.5	9960000	16.53	0.16	
12	8.27	36500000	15.15	0.19	
13	8.13	28690000	12.61	0.18	
14	9.29	62630000	15.36	0.14	
15	9.1	268460000	16.86	0.22	
16	8.95	6940000	18.93	0.16	
17	8.79	4890000	17.53	0.15	
18	9.87	860000	19.92	0.15	
19	9.79	50330000	19.21	0.18	
20	9.55	70100000	17.96	0.2	
21	9.58	4620000	18.51	0.18	

Figure 2.2 Dataset in excel sheet 1.2



	J	K	L	M	
1	Electricity from renewables (TWh)	Low-carbon electricity (% electricity)	Primary energy consumption per capita (kWh/person)	Energy intensity level of primary energy (MJ/$2017 PPP GDP)	Value_co2_emis
2	0.31	65.95744	302.59482	1.64	
3	0.5	84.745766	236.89185	1.74	
4	0.56	81.159424	210.86215	1.4	
5	0.63	67.02128	229.96822	1.4	
6	0.56	62.92135	204.23125	1.2	
7	0.59	63.440857	252.06912	1.41	
8	0.64	76.190475	304.4209	1.5	
9	0.75	78.94737	354.2799	1.53	
10	0.54	73.9726	607.8335	1.94	
11	0.78	82.97872	975.04816	2.25	
12	0.75	79.78723	1182.892	2.46	
13	0.6	76.92309	1436.1143	3.23	
14	0.74	84.09091	1324.1211	2.61	
15	0.89	80.180176	1060.7926	2.46	
16	1	86.2069	868.5762	2.25	
17	1.03	87.28814	970.0803	2.37	
18	1.06	87.603294	862.79114	2.24	
19	1.09	85.826775	829.31195	2.3	
20	0.97	82.90598	924.25085	2.44	
21	0.89	83.17757	802.61255	2.41	
22	0.68	85	702.888		
23	4.55	97.01493	9029.4375	4.13	
24	3.52	96.438354	8635.532	3.89	
25	3.48	95.60439	9443.555	4.1	
26	5.12	98.0843	10756.612	3.8	
27	5.41	97.65343	11586.951	3.96	
28	5.32	98.7013	11545.616	3.75	

Figure 2.3 Dataset in excel sheet 1.3



C7 25.390894

	N	O	P	Q	R	S	T	U
1	Value_co2_emissions_kt_by_country	Renewables (% equivalent primary energy)	gdp_growth	gdp_per_capita	Density\n(P/Km2)	Land Area(Km2)	Latitude	Longitude
2	760				60	652230	33.93911	67.709953
3	730				60	652230	33.93911	67.709953
4	1029.999971			179.4265792	60	652230	33.93911	67.709953
5	1220.000029		8.832277813	190.6838143	60	652230	33.93911	67.709953
6	1029.999971		1.414117981	211.3820742	60	652230	33.93911	67.709953
7	1549.999952		11.22971482	242.0313132	60	652230	33.93911	67.709953
8	1759.99999		5.357403251	263.7336019	60	652230	33.93911	67.709953
9	1769.999981		13.82631955	359.6931579	60	652230	33.93911	67.709953
10	3559.999943		3.924983822	364.663542	60	652230	33.93911	67.709953
11	4880.000114		21.39052839	437.2687402	60	652230	33.93911	67.709953
12	7110.000134		14.36244147	543.3065262	60	652230	33.93911	67.709953
13	8930.000305		0.426354785	591.1900302	60	652230	33.93911	67.709953
14	8079.999924		12.75228709	638.8458516	60	652230	33.93911	67.709953
15	5989.999771		5.600744658	624.3154545	60	652230	33.93911	67.709953
16	4880.000114		2.724543364	614.2233424	60	652230	33.93911	67.709953
17	5949.999809		1.45131466	556.0072209	60	652230	33.93911	67.709953
18	5300.000191		2.260314201	512.0127781	60	652230	33.93911	67.709953
19	4780.00021		2.647003202	516.6798622	60	652230	33.93911	67.709953
20	6070.000172		1.189228128	485.6684187	60	652230	33.93911	67.709953
21	6079.999924		3.911603419	494.1793499	60	652230	33.93911	67.709953
22			-2.351100673	516.7478708	60	652230	33.93911	67.709953
23	3170		6.946216585	1126.68334	105	28748	41.153332	20.168331
24	3230		8.293312636	1281.659826	105	28748	41.153332	20.168331
25	3759.99999		4.536524157	1425.124219	105	28748	41.153332	20.168331
26	4070.000172		5.528637464	1846.120121	105	28748	41.153332	20.168331
27	4250		5.51466791	2373.581292	105	28748	41.153332	20.168331
28	4030.00021		5.526424241	2673.786584	105	28748	41.153332	20.168331

global-data-on-sustainable-ener

Figure 2.4 Dataset in excel sheet 1.4

A. TITLE

Sales analysis using Power-BI

B. ATTRIBUTES IN THE DATASET

Table 2.1 Description of the dataset

Index	Column Name	Description
1.	Entity	This column contains a Country Name
2.	Year	This column provides Date of year
3.	Access to electricity (% of population)	Percentage of the population with access to electricity.
4.	Access to clean fuels for cooking	Percentage of the population with access to clean cooking fuels
5.	Renewable-electricity-generating-capacity-per-capita	Percapita renewable electricity generating capacity
6.	Financial flows to developing countries (US $)	Financial flows to developing countries in US dollars
7.	Renewable energy share in the total final energy consumption (%)	Percentage of renewable energy in total final energy consumption
8.	Electricity from fossil fuels (TWh)	This column likely represents the Electricity generated from fossil fuels in terawatt-hours
9.	Electricity from nuclear (TWh)	Electricity generated from nuclear sources in terawatt-hours
10.	Electricity from renewables (TWh)	Electricity generated from renewable sources in terawatt-hours
11.	Low-carbon electricity (% electricity)	Percentage of electricity generated from low-carbon sources
12.	Primary energy consumption per capita (kWh/person)	Per capita primary energy consumption in kilowatt-hours

13.	Energy intensity level of primary energy (MJ/$2017 PPP GDP)	Energy intensity level of primary energy in megajoules per constant 2017 PPP GDP dollar
14.	Value_co2_emissions_kt_by_country	Carbon dioxide emissions in kilotons
15.	Renewables (% equivalent primary energy)	Percentage of renewables in equivalent primary energy
16.	gdp_growth	GDP growth rate
17.	gdp_per_capita	GDP per capita
18.	Density (P/Km2)	Population density in persons per square kilometer
19.	Land Area(Km2)	Total land area in square kilometers
20.	Latitude	Geographic latitude
21.	Longitude	Geographic longitude

2.3 BI Questions

This analysis is basically concerned with several issues such as:

How has the percentage of population with access to electricity evolved globally over 2000-2020? (Line chart)

- What is the trend in the percentage of the population with access to clean cooking fuels?
- How does renewable electricity generating capacity per capita vary across different countries?

- What is the pattern of financial flows to developing countries for sustainable energy projects?
- How has the percentage of renewable energy in total final energy consumption changed globally?
- What is the contribution of electricity generated from fossil fuels over the years?
- How does nuclear electricity generation vary across different countries?
- Which countries lead in electricity generation from renewable sources?
- What is the global proportion of low-carbon electricity in the total electricity generated?
- How does per capita primary energy consumption vary among different countries?
- How does the energy intensity level correlate with GDP growth?
- What is the trend in carbon dioxide emissions globally over the years?
- How has the percentage of renewables in equivalent primary energy changed over time?
- Is there a correlation between GDP growth and the adoption of sustainable energy practices?
- How does GDP per capita correlate with access to sustainable energy?
- Does population density influence the adoption of sustainable energy practices?
- What is the distribution of land area among countries with different levels of renewable energy adoption?
- How does geographic location influence the distribution of renewable energy sources?

2.4 Data loading

Click the "Home" option to begin the process of loading data into Microsoft Power BI for the "Energizing the Future" sustainable energy dataset. Select "Text/CSV file" as the file format from the menu that appears after selecting the "Get Data" option. Next, find and open the particular CSV file that has the data on sustainable energy. To begin the import procedure, click "Open" to confirm the selection. This simplified process guarantees the dataset's smooth integration, opening the door for in-depth analysis and visualization in the Power BI environment.

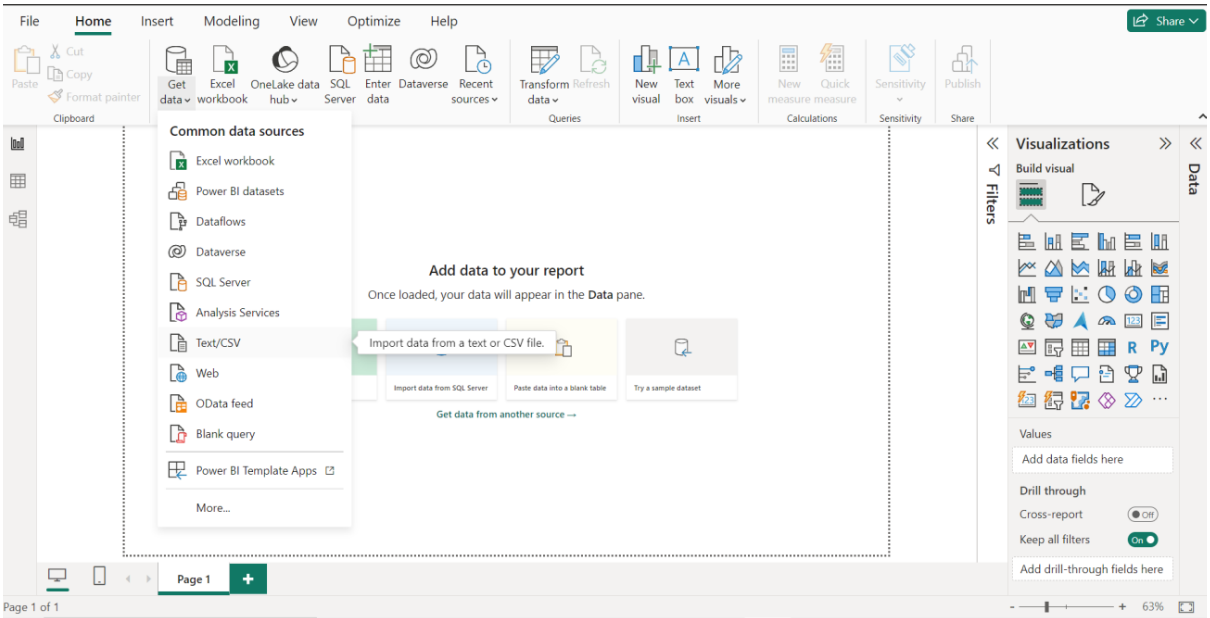


Figure 2.5 Importing the CSV file

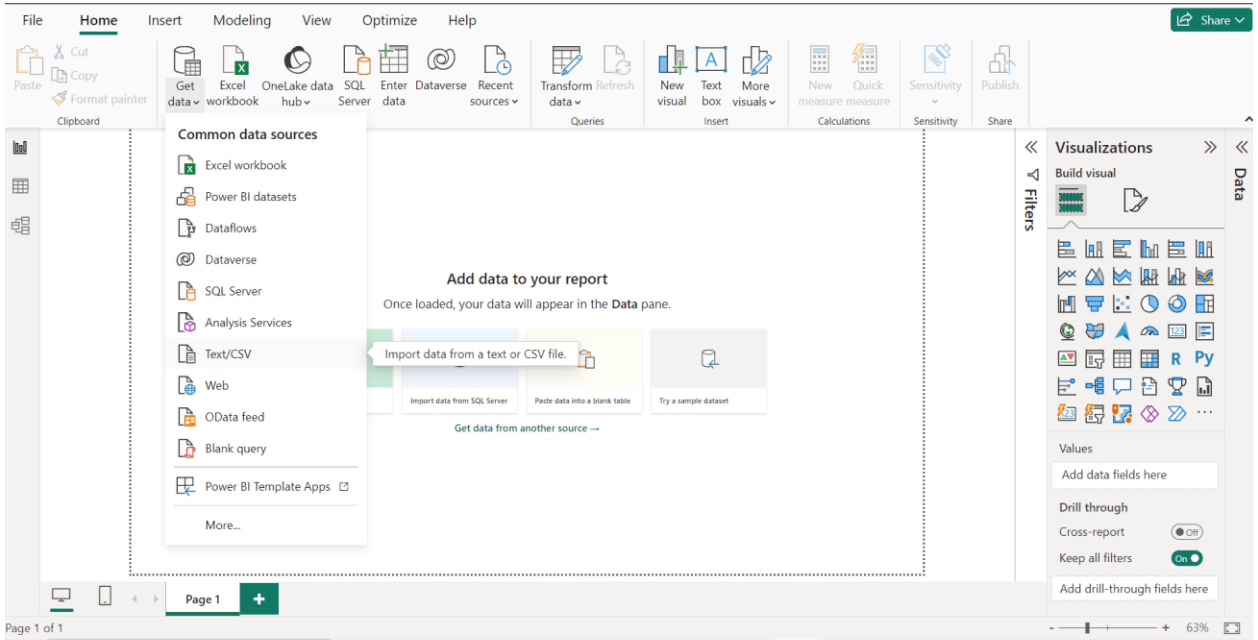


Figure 2.6 Transforming Imported Data

2.5 Data cleaning

Once the data has been transformed, the next step is to validate its accuracy. Subsequently, it is crucial to examine the primary dataset for potential errors or occurrences of missing values.

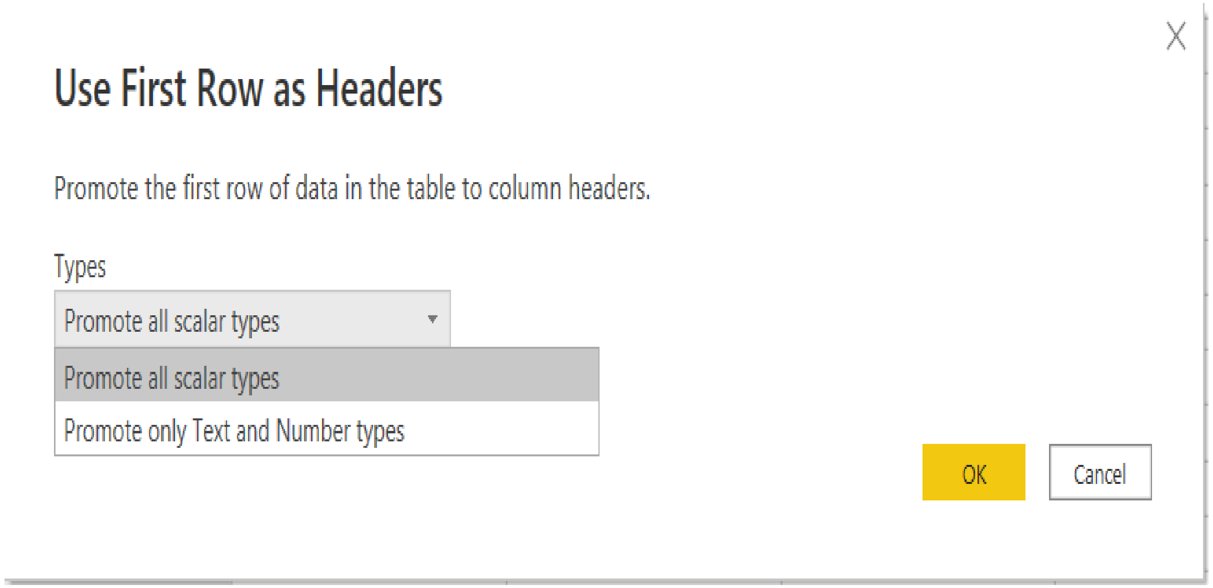


Figure 2.7 promoting the first row as headers

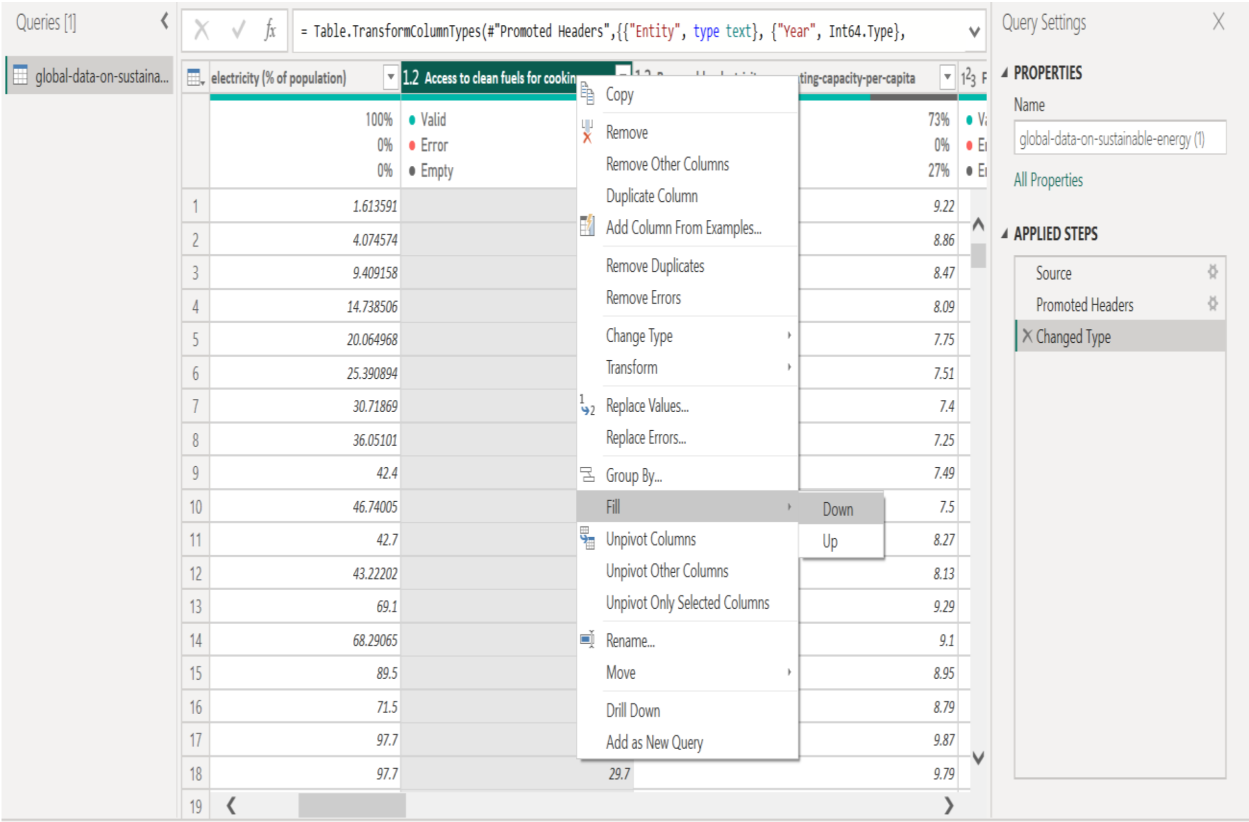


Figure 2.8 Filling Empty Values

2.6 Data Handling

Handling the sustainable energy dataset requires a meticulous approach for extracting meaningful insights. Initially, importing the data into tools like Microsoft Power BI facilitates further analysis. Transformation processes are then applied to ensure alignment with analytical requirements, while validity checks address accuracy and outliers for enhanced reliability. Managing missing values, through imputation or exclusion, is integral to analysis needs. Employing Power BI, visualization becomes key for effective communication, utilizing charts and dashboards to represent trends and comparisons. Statistical analysis, including averages and correlations, deepens insights, while ongoing quality assurance ensures data integrity and validation against expectations.

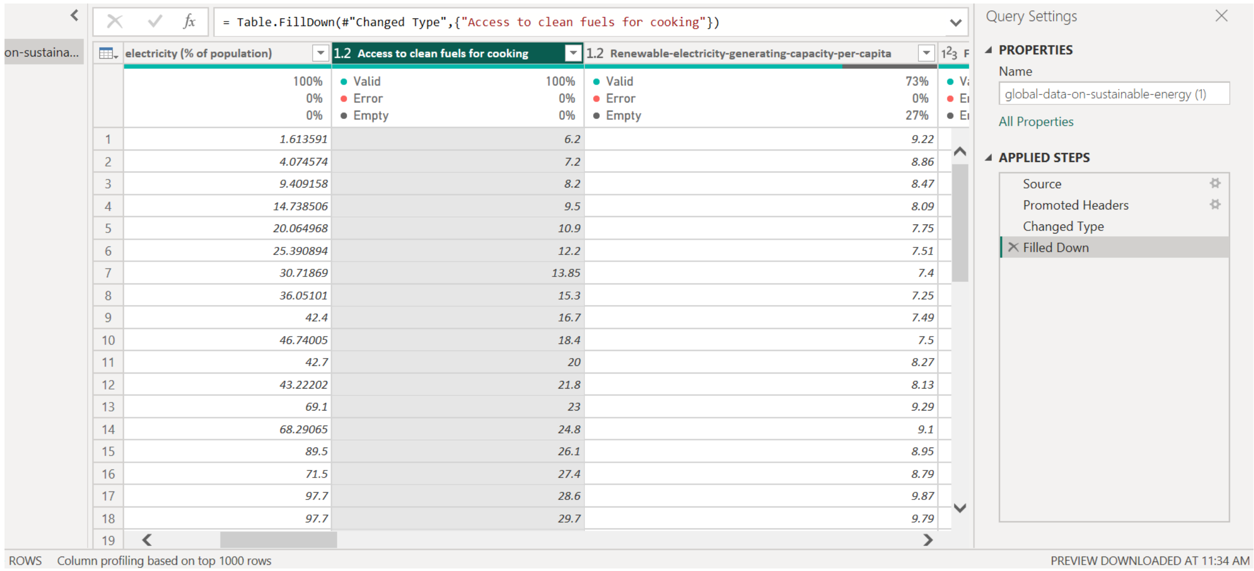


Figure 2.9 Filled Empty With Down

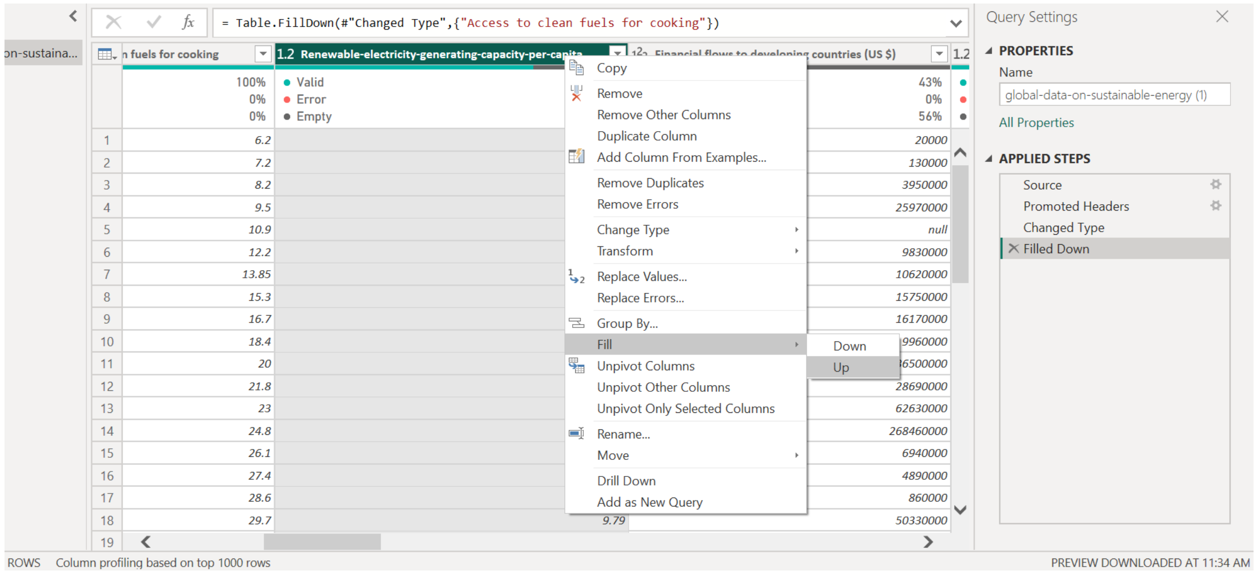


Figure 2.10 Filing Empty Rows

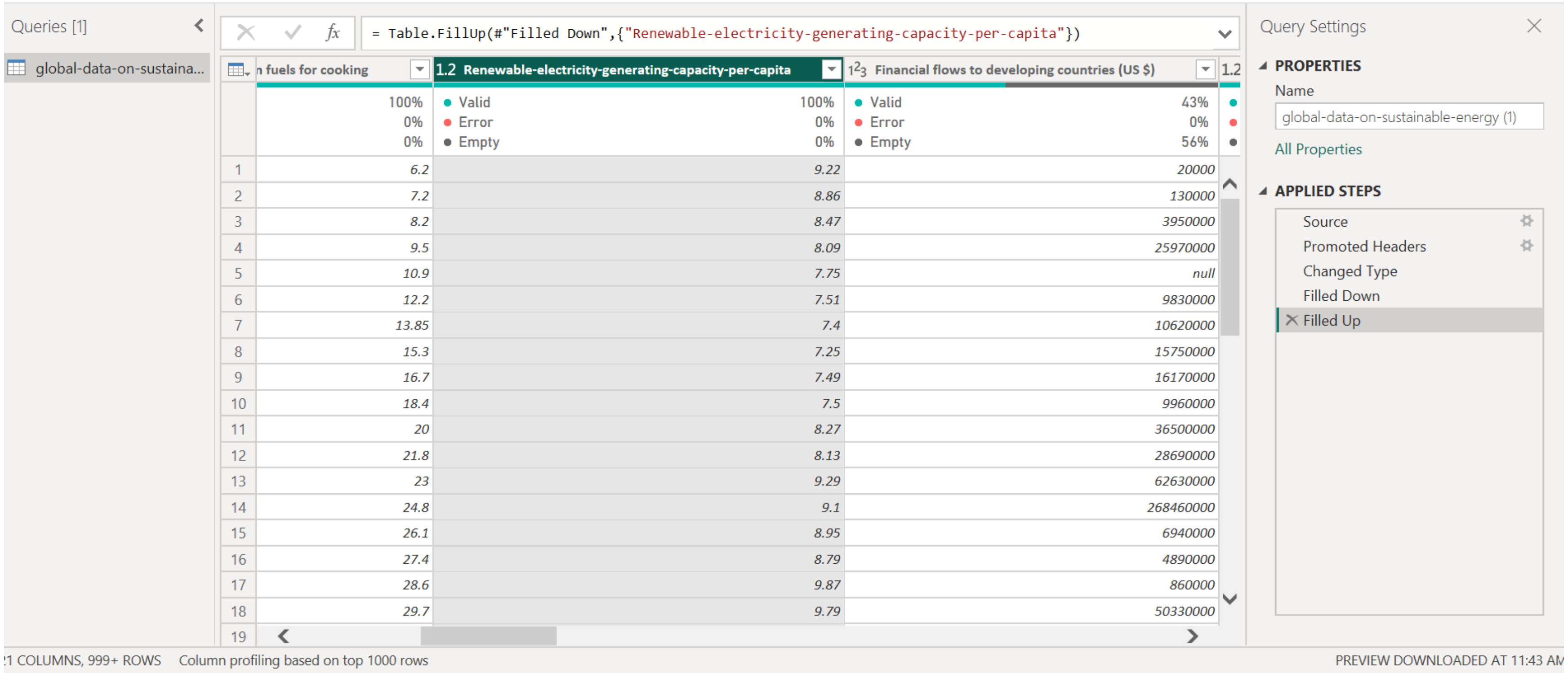


Figure 2.11 Filing Empty with Upper

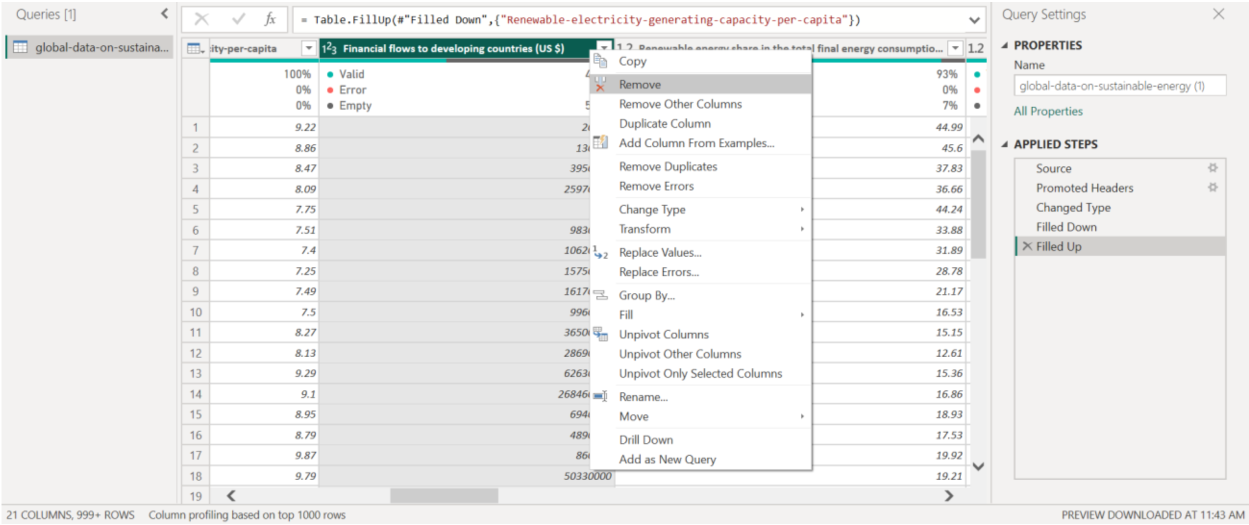


Figure 2.12 Removing Column

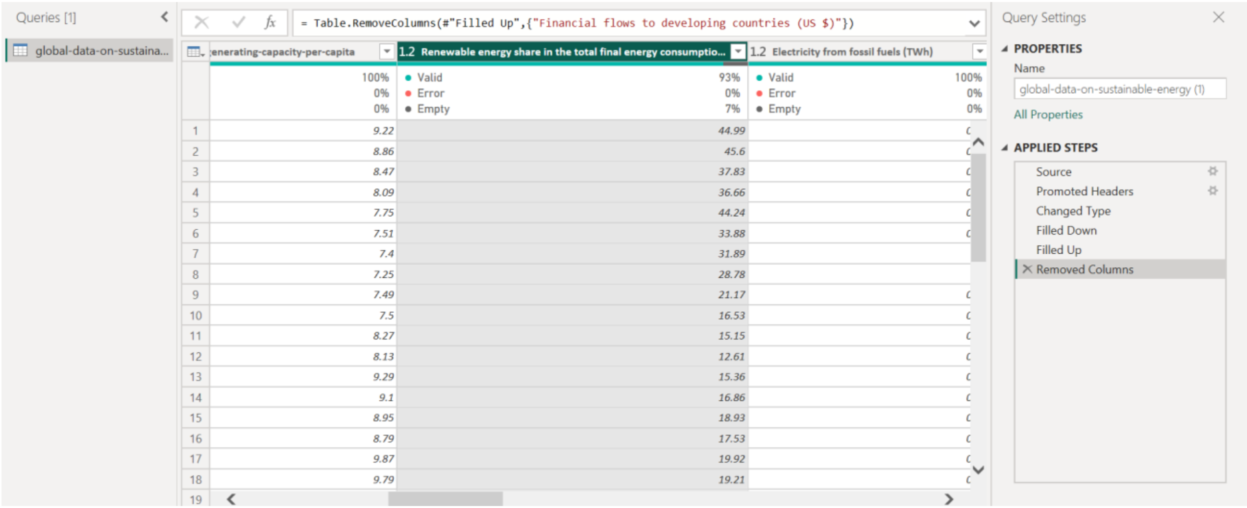


Figure 2.13 After Removing Column

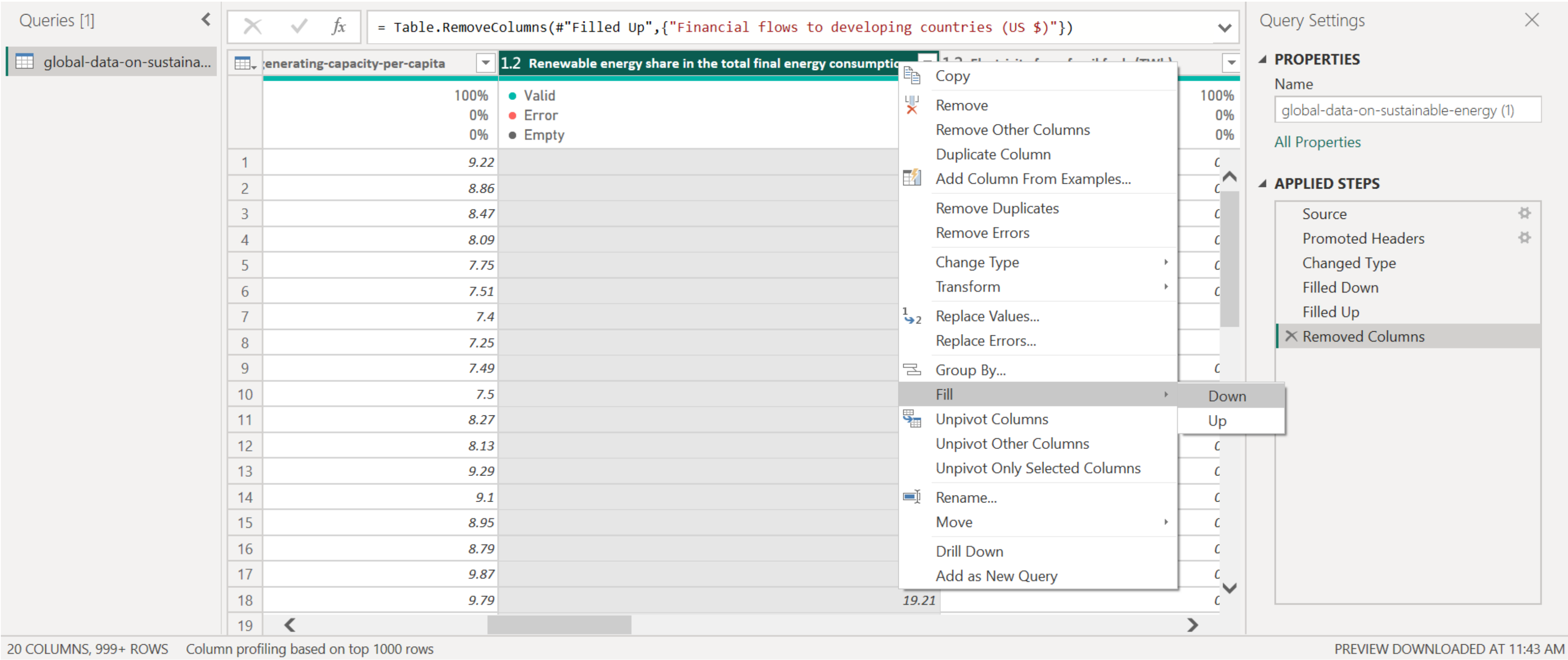


Figure 2.14 White Space Filling with Down Value

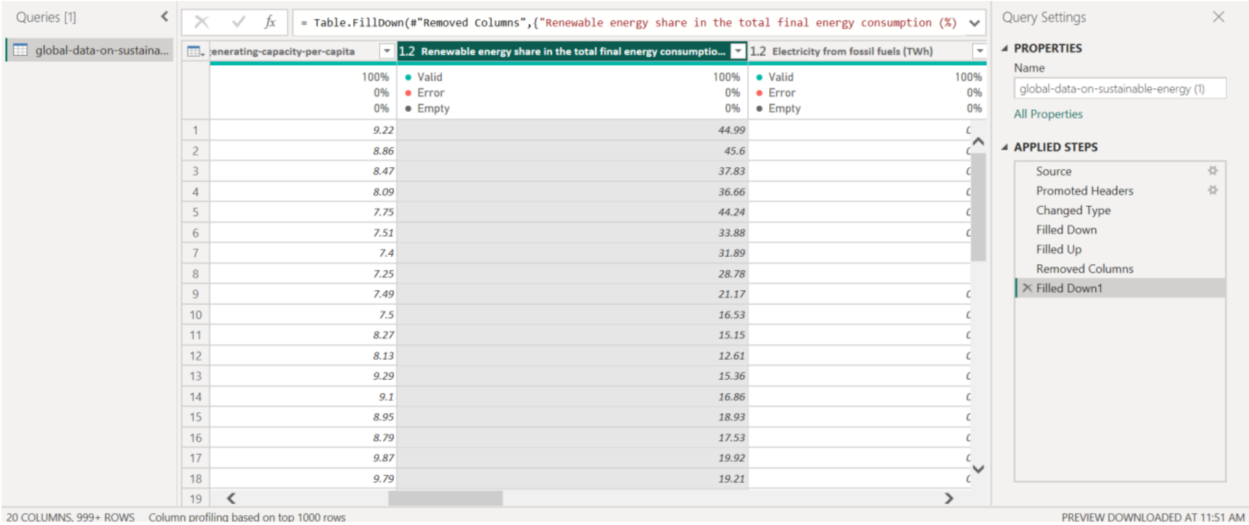


Figure 2.15 Filled white space with down values

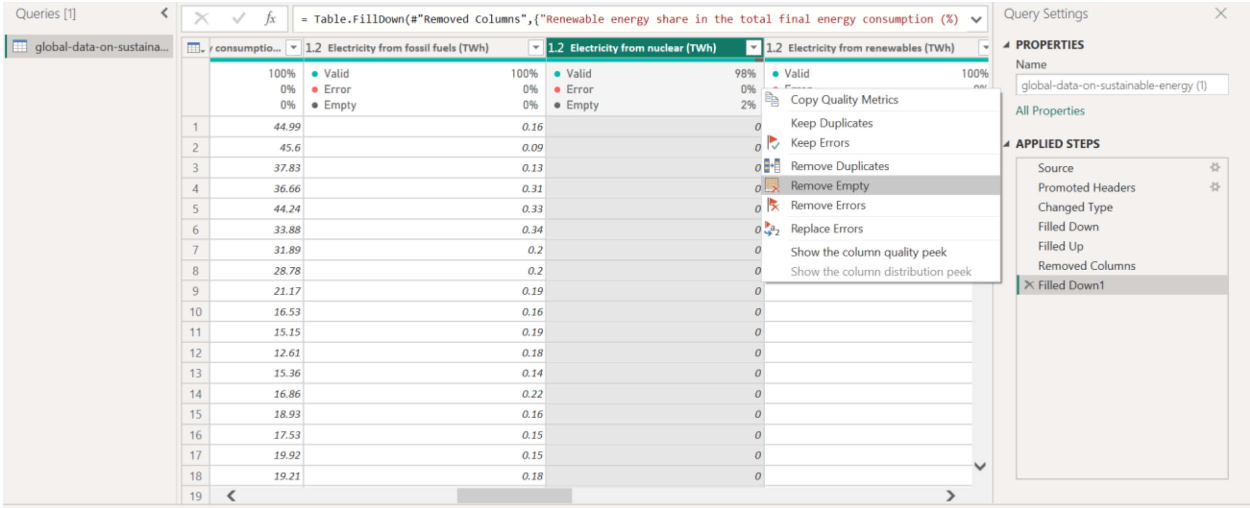


Figure 2.16 Removing empty rows

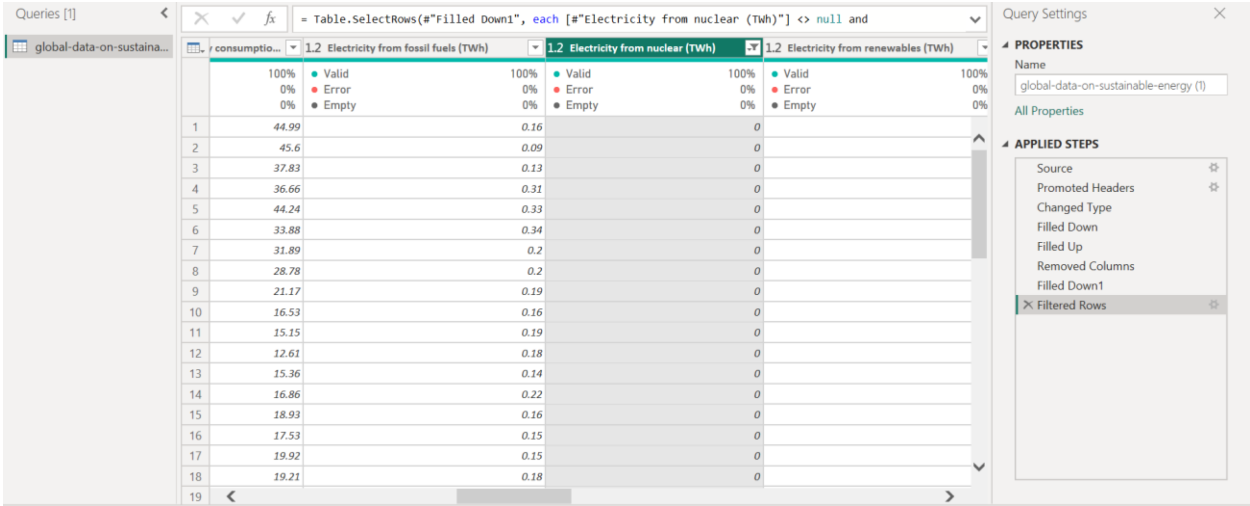


Figure 2.17 Filtered row

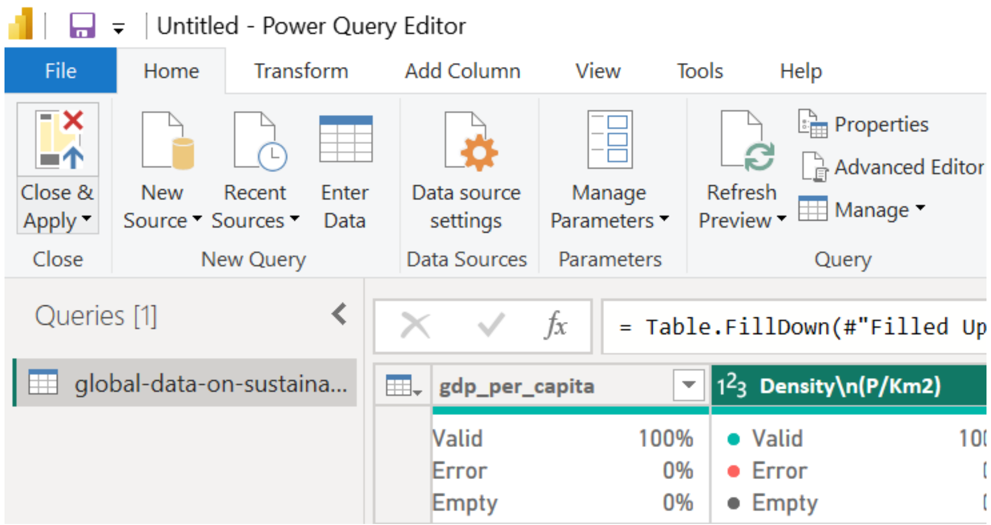


Figure 2.18 Applying all changes

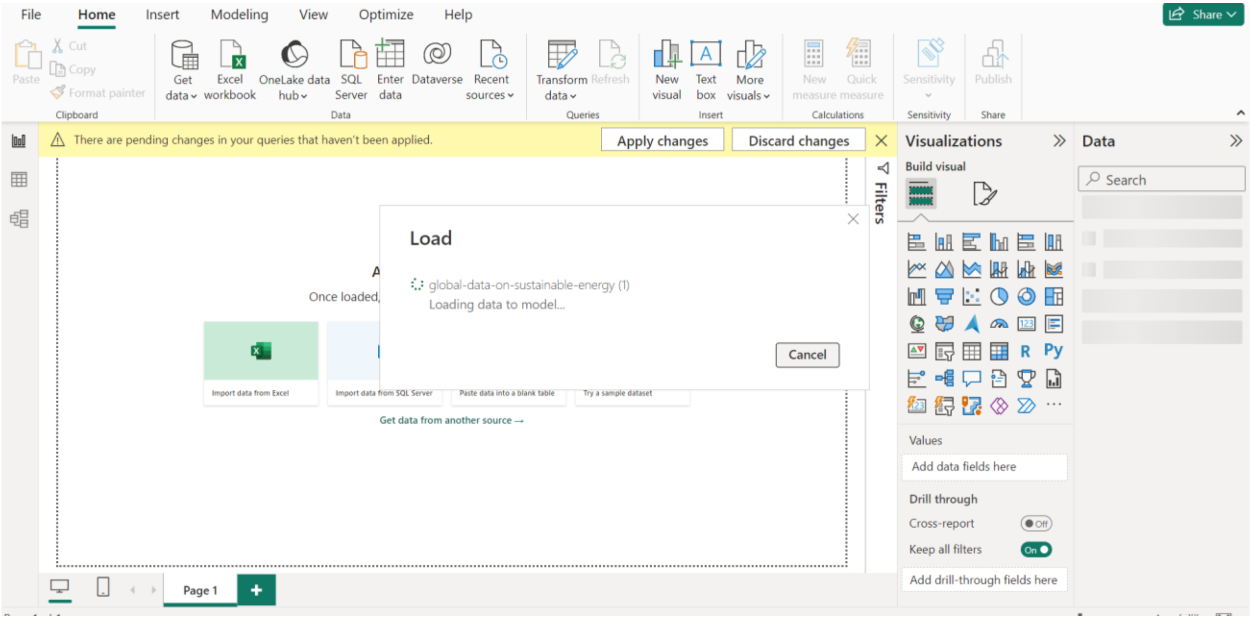


Figure 2.19 loading data

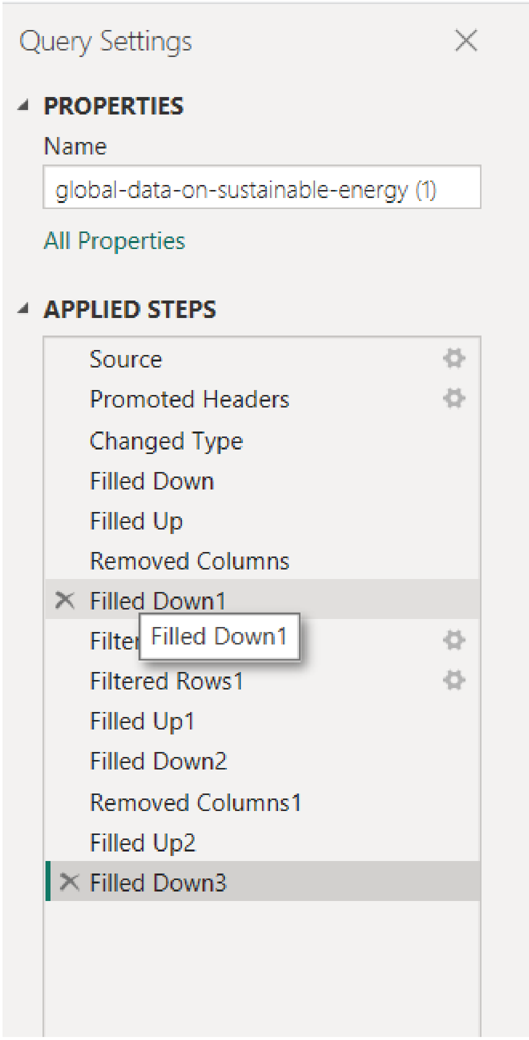


Figure 2.20 Cleaning the main dataset and the steps applied are shown

2.7 A Data Modelling and Relationship between tables

Following the completion of the task mentioned earlier, the next step is to create connections between the tables, as demonstrated in the accompanying figure.

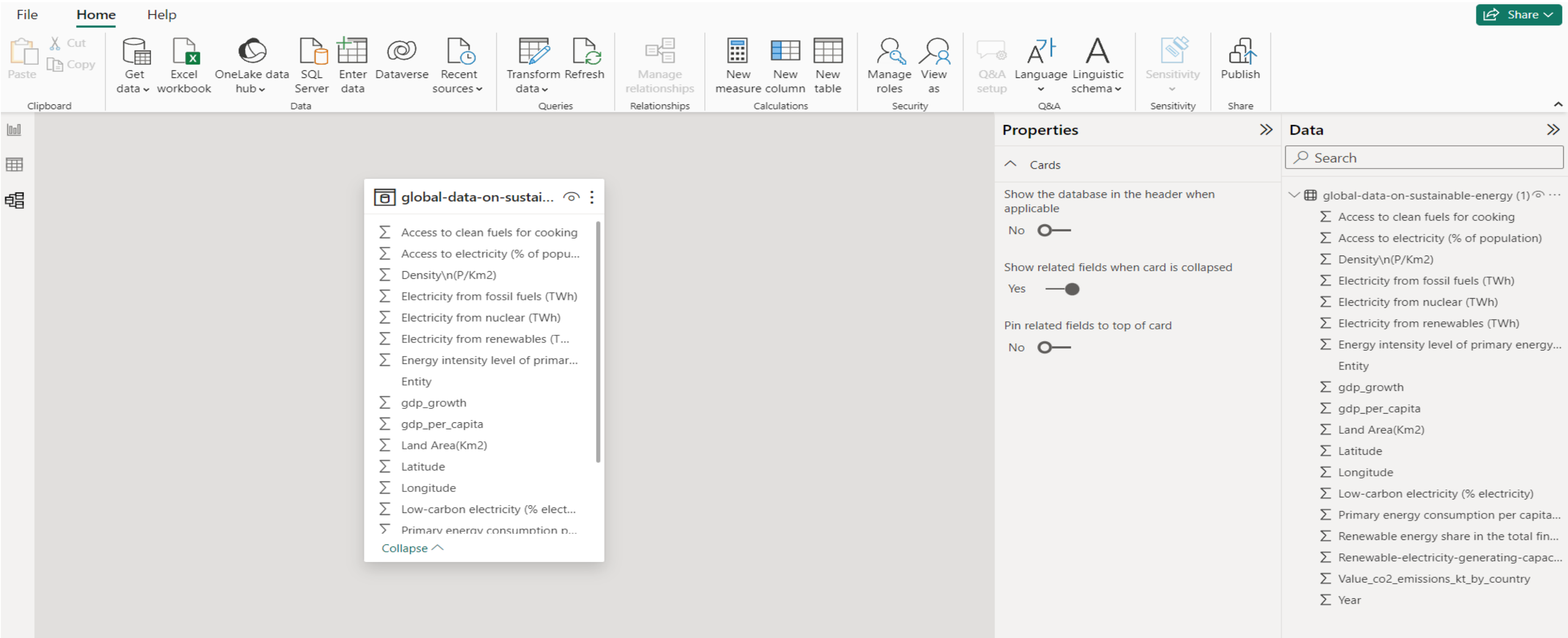


Figure 2.21 Data modelling and relationship between tables.

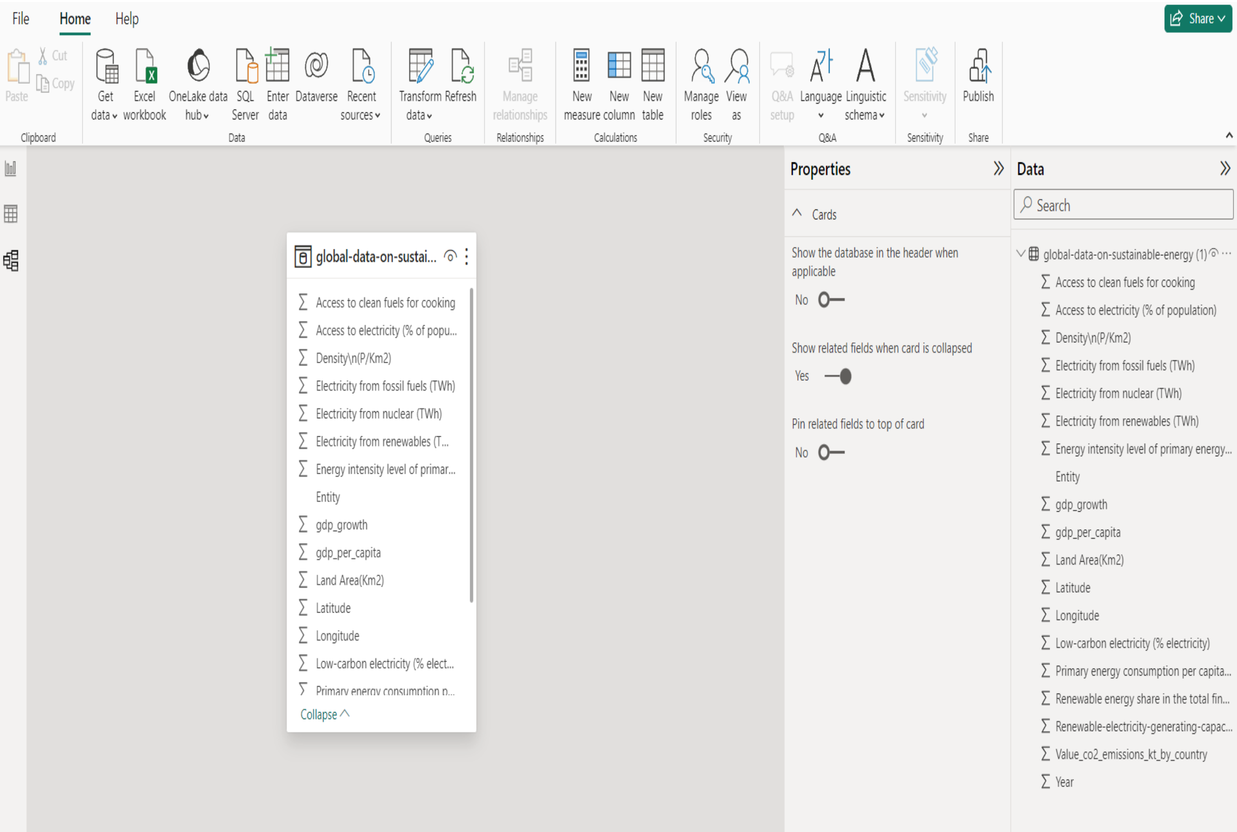


Figure 2.22 Relationship between tables.

The relationships between the tables can be formed easily, and the procedure for it is shown in the figure below.

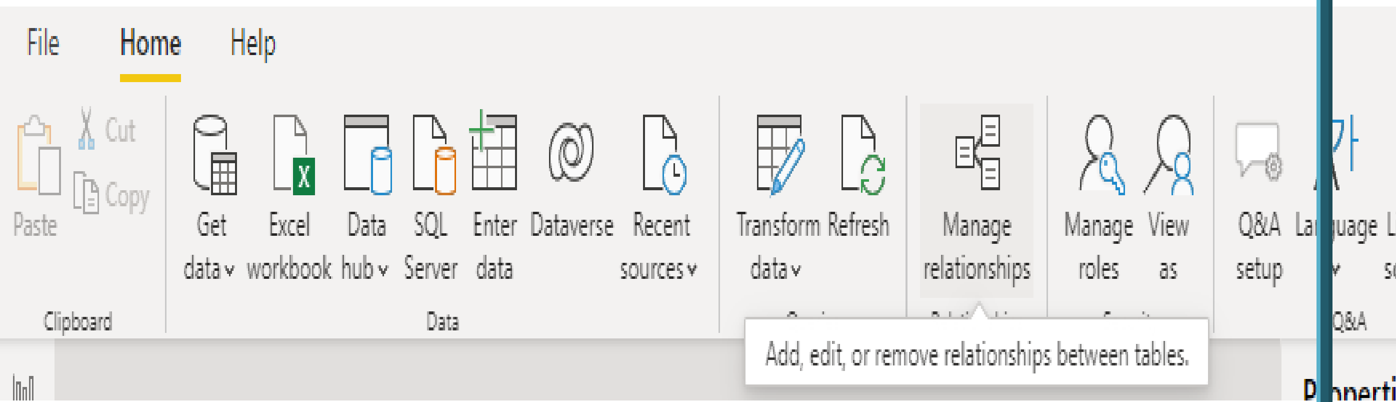


Figure 2.23 Select manage relationships to manage relationships between tables.

2.8 Data Analysis Expressions (DAX) Functions

Data Analysis Expressions (DAX) serves as a powerful tool in Power BI for deriving insights from the sustainable energy dataset. It enables the creation of calculated columns, custom tables, and measures. To generate a custom column, navigate to "Add Column" and select "Custom Column" as illustrated. Follow the outlined steps in the accompanying figures to apply DAX functions, allowing for the calculation and delivery of values crucial for in-depth analysis of sustainable energy trends and metrics.

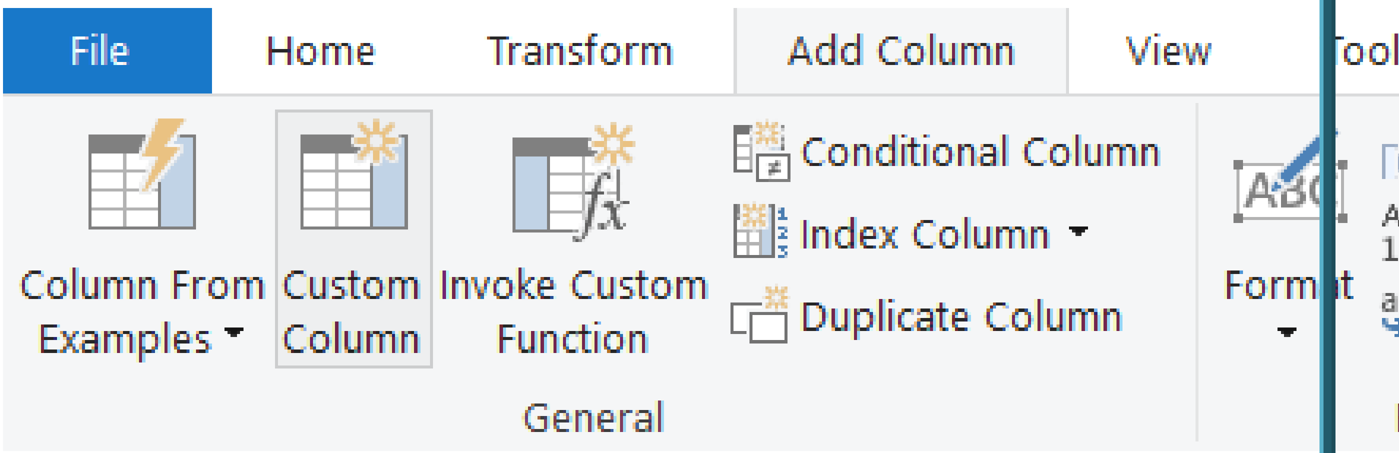


Figure 2.24 Adding of custom column

Now apply the changes by clicking on the file and in that select Apply, as shown below.

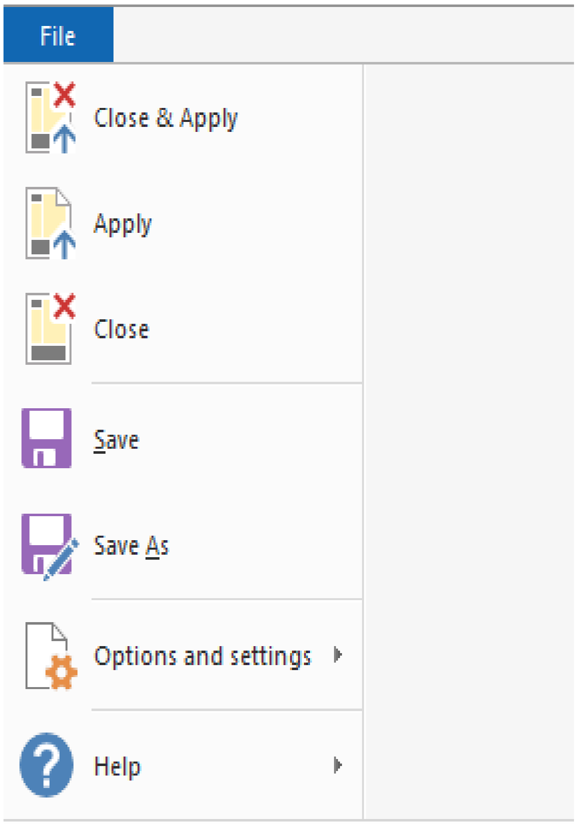


Figure 2.25 Apply the changes by clicking on Apply option.

Sustainable Energy Trends Data Analysis

Section C

Business Intelligence Solutions

3 Sustainable Energy Trends data analysis

3.1 Introduction

The introduction to sustainable energy data analysis lays the foundation for an in-depth exploration of the extensive and nuanced information encapsulated within datasets pertaining to sustainable energy trends from 2000 to 2020. Enriched with details about access to electricity, renewable capacities, financial flows, and environmental metrics, this dataset presents a valuable opportunity to extract meaningful insights. Through careful analysis, one can reveal patterns, correlations, and trends that foster a comprehensive understanding of the dynamics shaping the global landscape of sustainable energy practices.

3.2 Data Model

The data model is carried out in Section 1 and the overall Model is

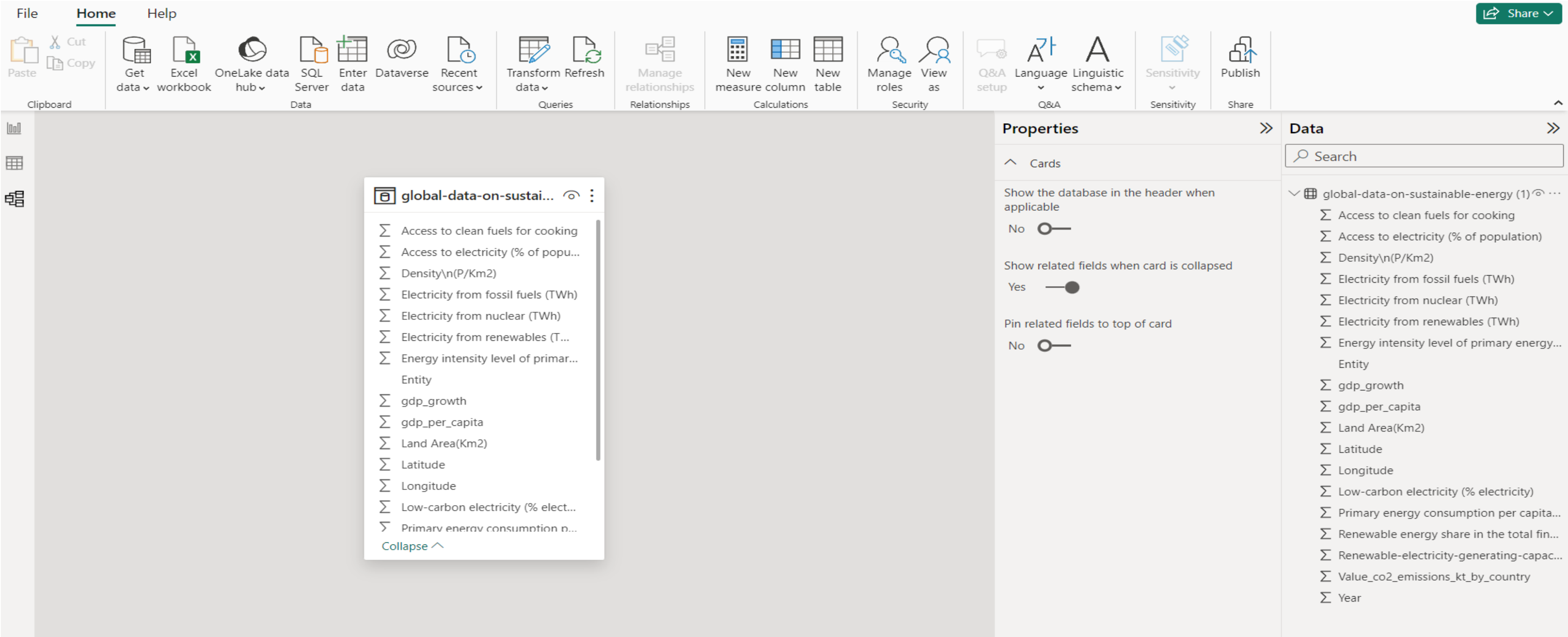


Figure 3.1 Final Data Model

Introduction to the visuals in the Power-BI

This section delves into diverse visualizations showcasing insights gleaned from the sustainable energy dataset. It emphasizes distinct chart types and elucidates the information encapsulated within these visuals, as illustrated in the figure below. With the completion of data cleaning and pre-processing, the focus shifts to applying various visualizations. The ensuing steps involve

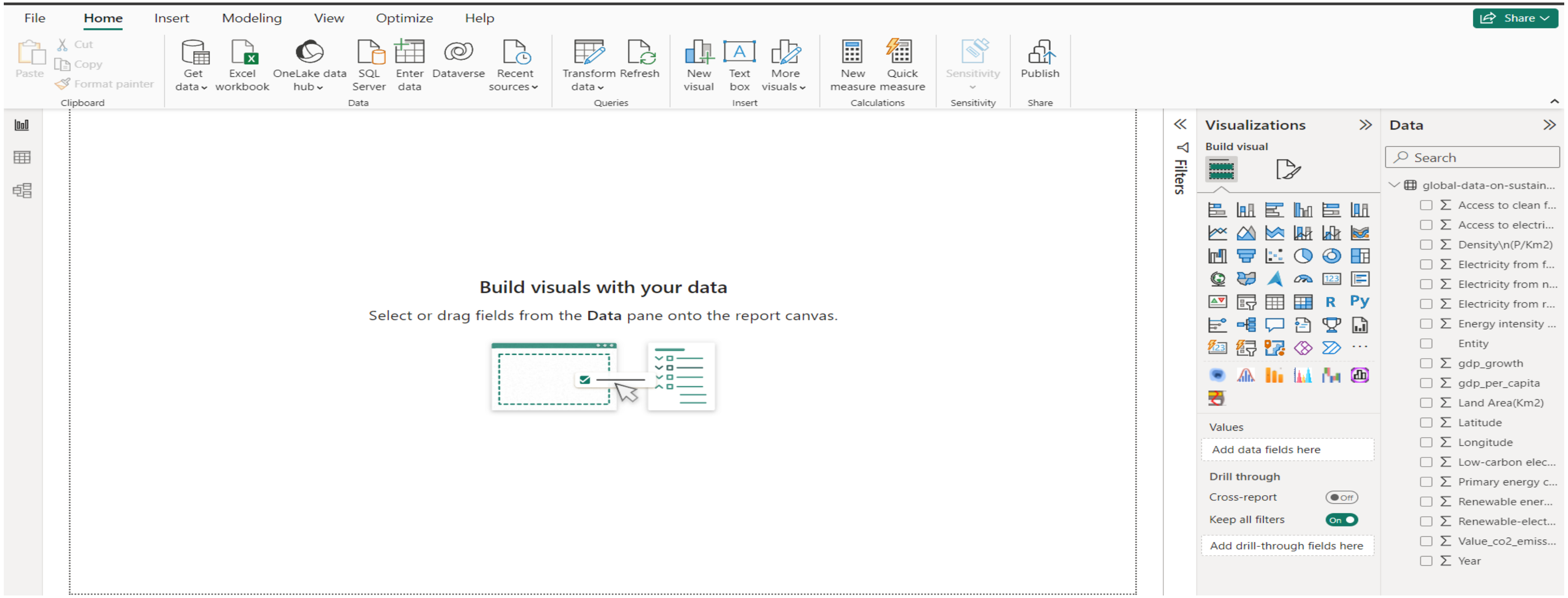


Figure 3.2 Introduction to visualizations

3.3 Visualization options

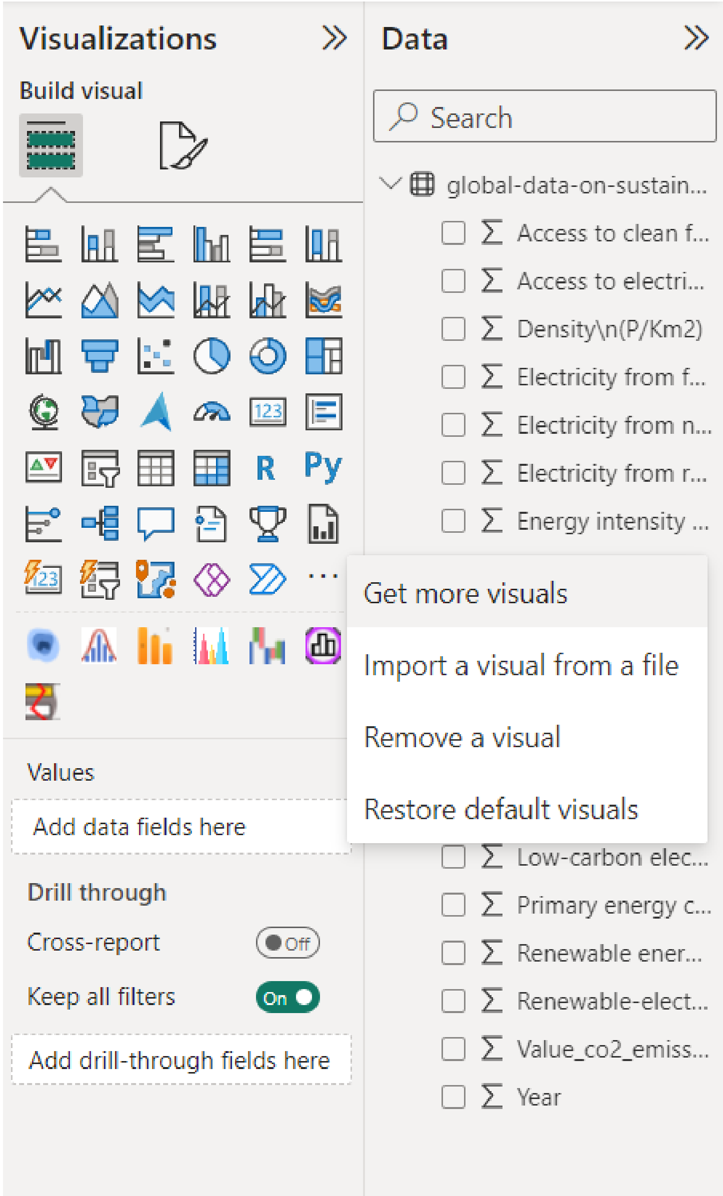


Figure 3.3 Get more Visuals

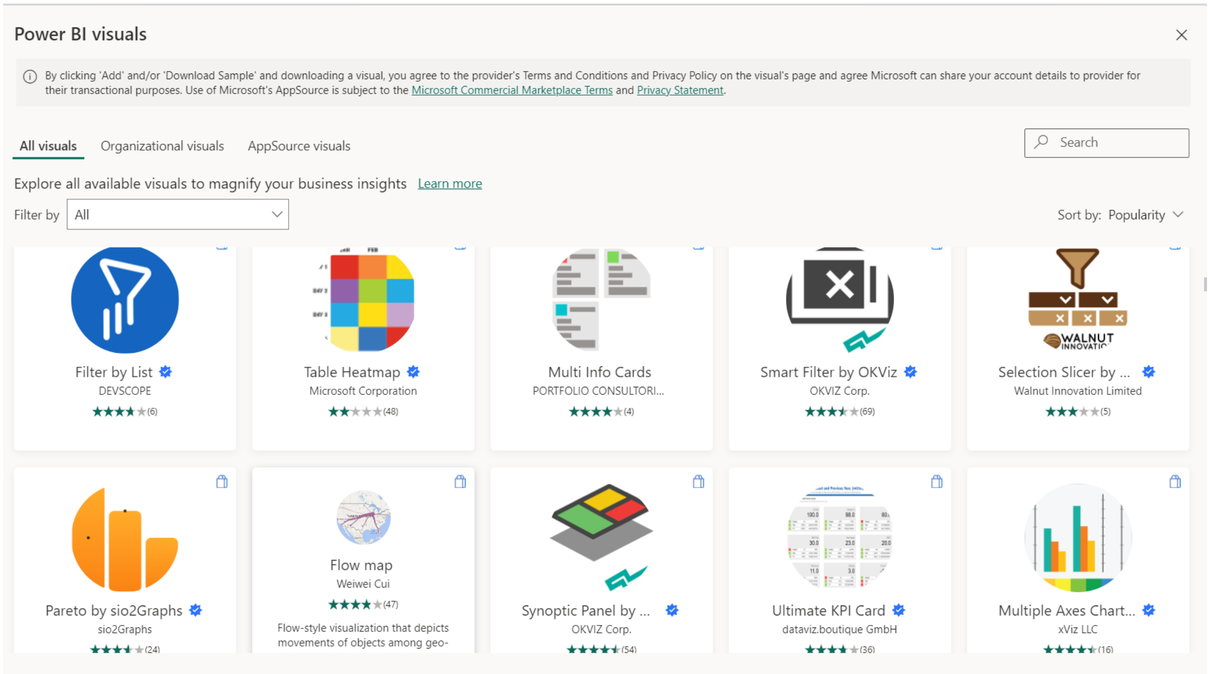


Figure 3.4 Searching for an animated chart

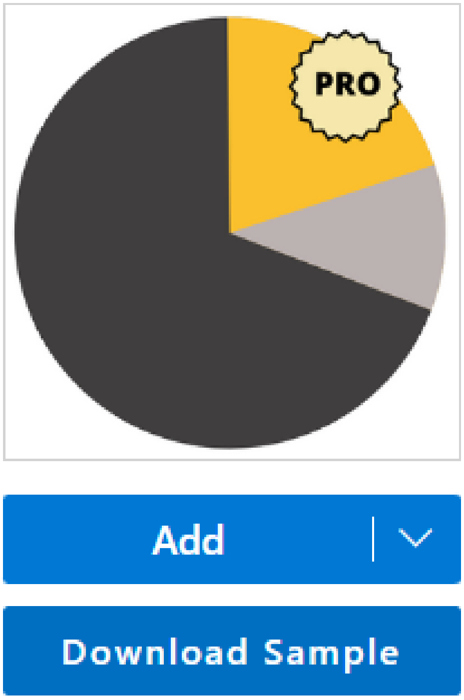


Figure 3.5 Adding the sunburn chart

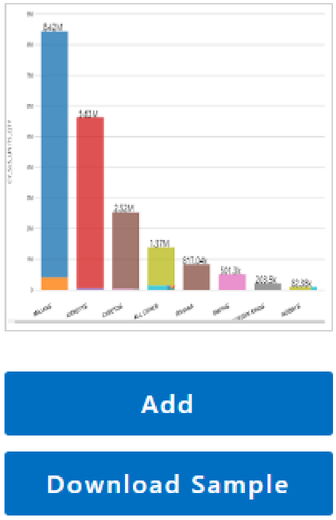


Figure 3.6 Adding the clustered stacked bar chart

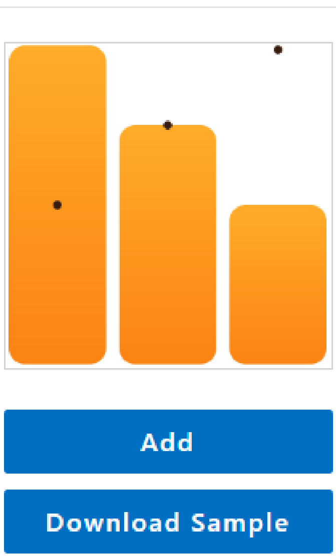


Figure 3.7 Adding the Sio2 Graph

3.4 Findings based on analysis.

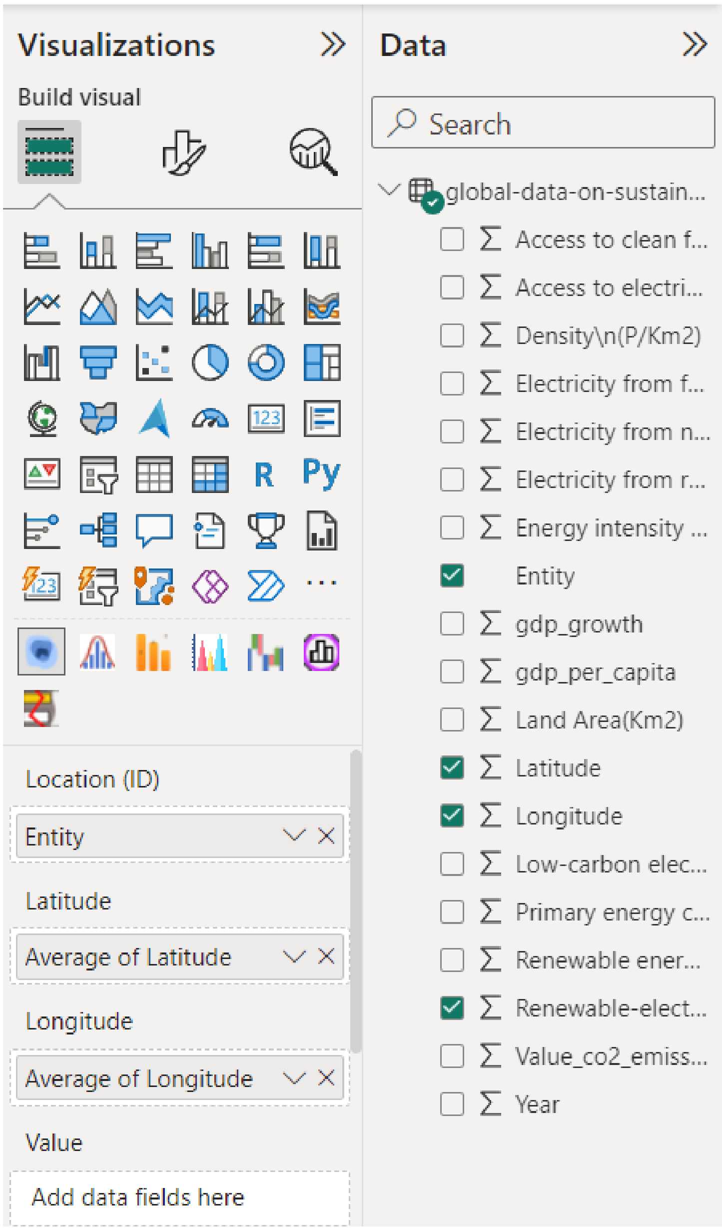


Figure 3.8 Preparing the visualize the Map plot

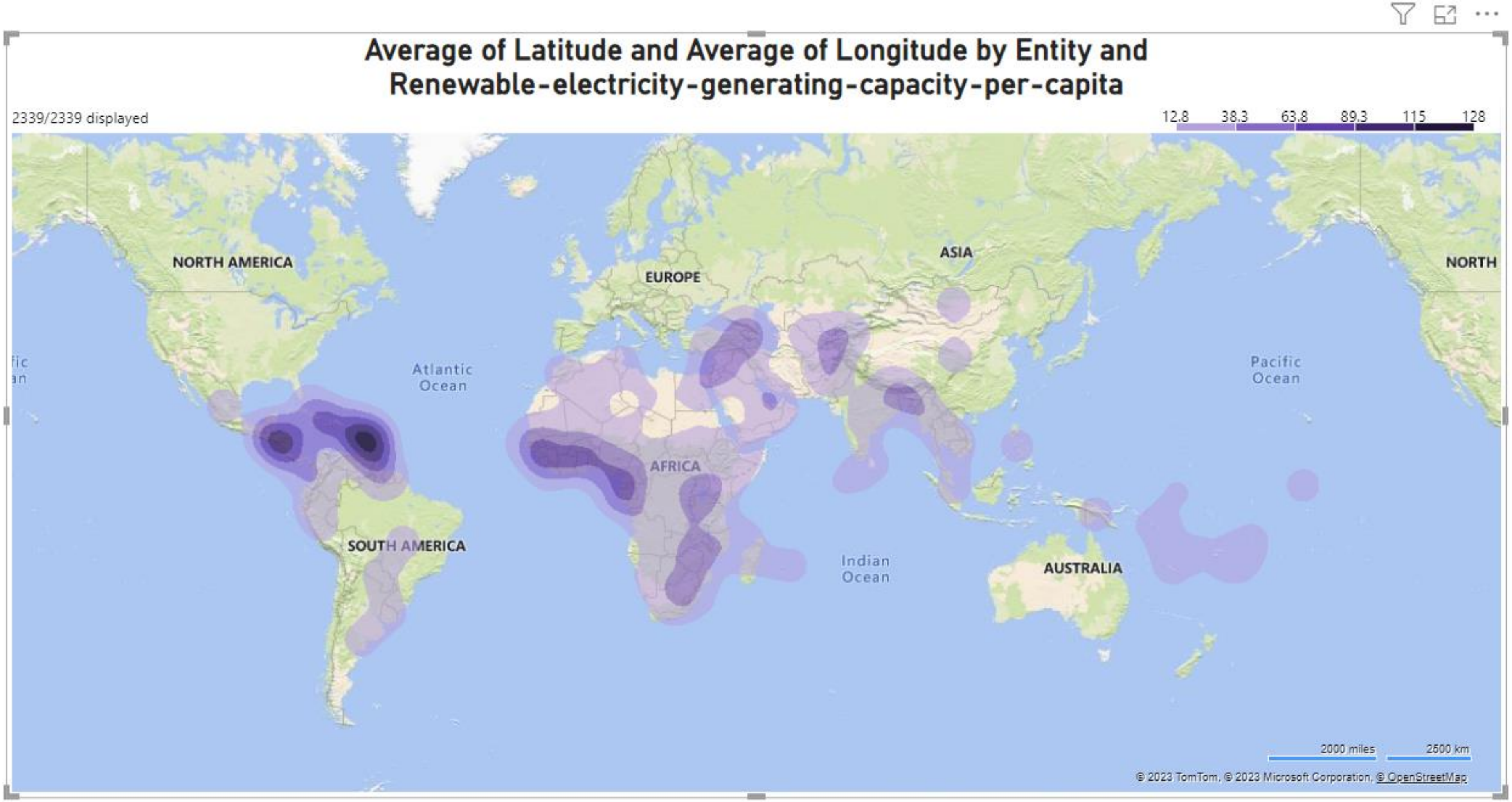


Figure 3.9 Renewables Light Up the High Latitudes: A Glimpse of the Geopolitics of Green Energy

This world map reveals a fascinating link between geography and renewable energy. Countries at higher latitudes, bathed in more sunlight and whipped by stronger winds, generally boast higher capacities for generating clean power. Canada, Norway, and Sweden, nestled in the embrace of the North, stand out as shining examples, their renewable energy potential fuelled by abundant hydropower and wind resources.

This glimpse into the geopolitics of renewable energy underscores the complex interplay between geography, policy, and resource availability. While nature has given high latitudes a head start, the race to a clean energy future is far from a foregone conclusion. As technology evolves and political will strengthens, the map of renewable energy may well be redrawn, with new players emerging from unexpected corners of the globe.

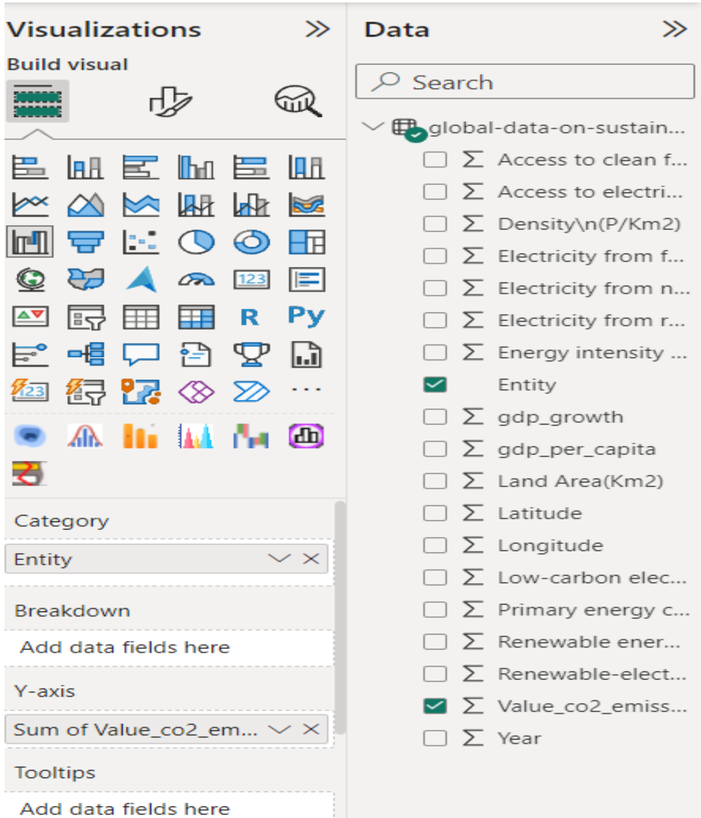


Figure 3.10 Selecting Water Fall Chart

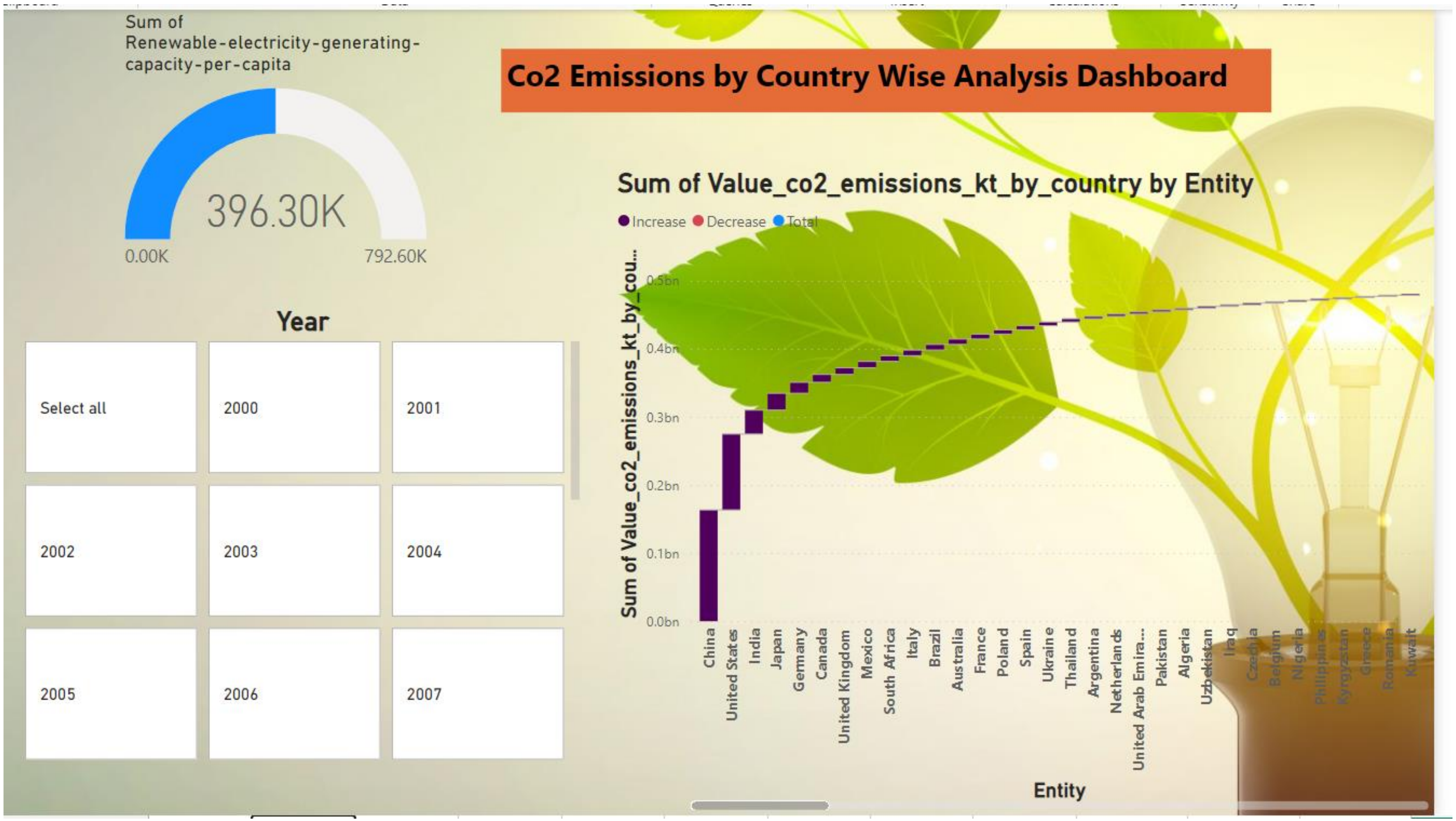


Figure 3.11 Co2 Emissions by Country wise Analysis Dasboard

- It is a graph of the renewable energy generation capacity-per-capita by country.
- The graph shows that the renewable energy generation capacity-per-capita has increased in recent years.
- The graph also shows that the renewable energy generation capacity-per-capita is higher in developed countries than in developing countries

The diagram presents a year-wise breakdown of medal counts, highlighting that information is available for the years 1980 to 2020. Notably, in the year 1992, the highest medal count was recorded, reaching 4,161.

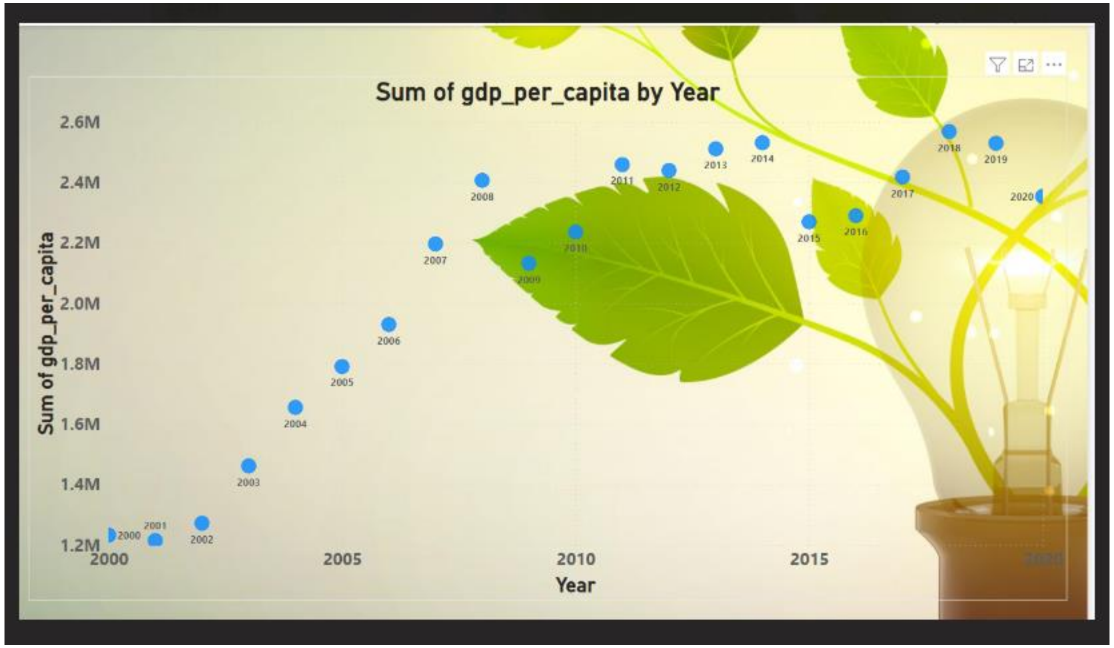


Figure 3.12 Sum of GDP percapita by Year

- The image shows a graph of the sum of GDP per capita by year.
- The graph shows that the sum of GDP per capita has increased steadily over time, from around $1.2 million in 2000 to around $2.6 million in 2020.
- The graph also shows some fluctuations in GDP per capita growth. For example, there was a sharp increase in GDP per capita growth between 2005 and 2010, followed by a period of slower growth between 2010 and 2015.
- It is important to note that the data in the graph is not adjusted for inflation. This means that the increase in GDP per capita may be partly due to rising prices.

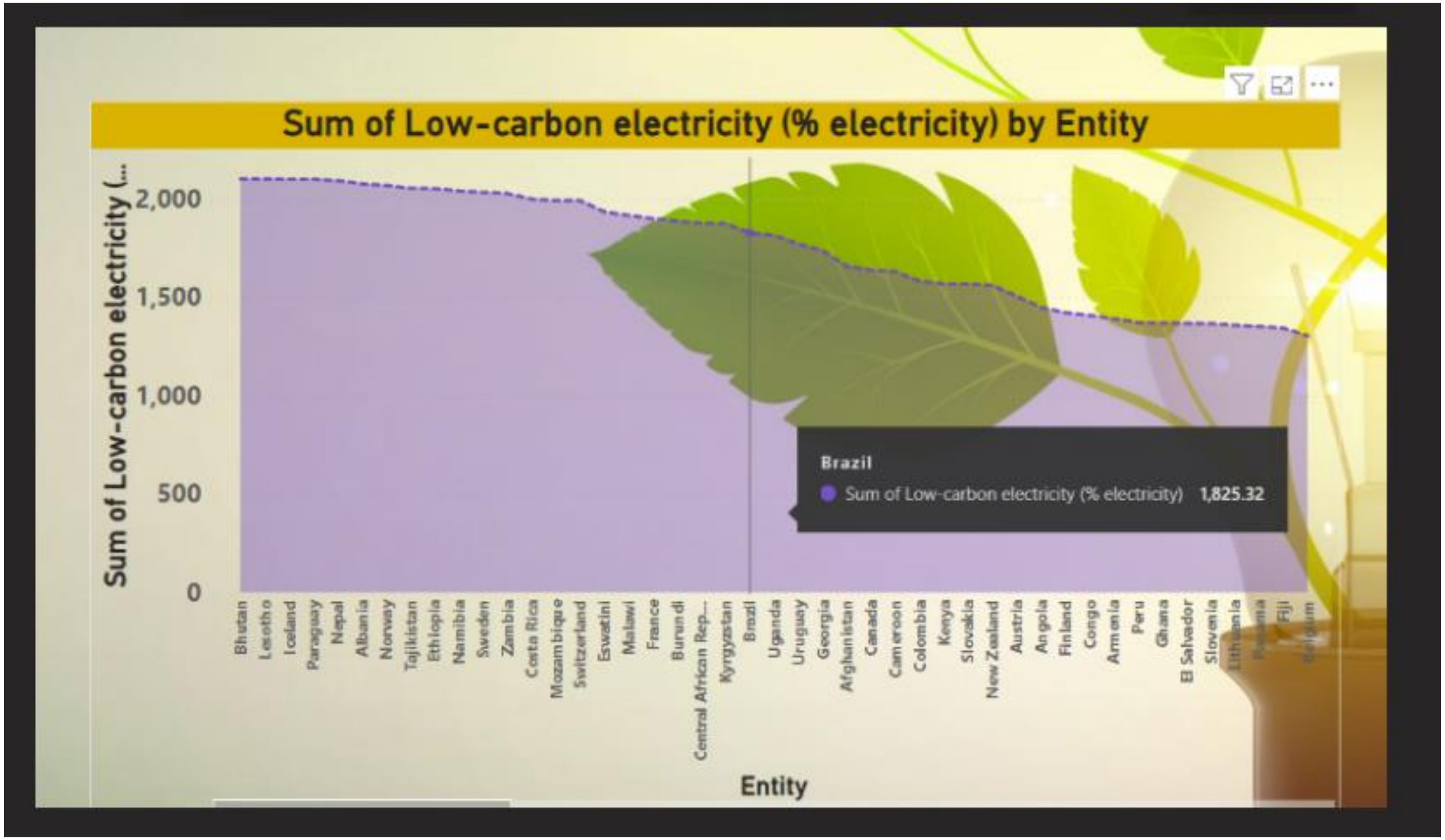


Figure 3.13 Low Carbon Electricity Country Wise

- The image shows a graph of the sum of low-carbon electricity, as a percentage of total electricity, by country.
- The graph shows that Brazil has the highest sum of low-carbon electricity, at 1,825.32%. This means that all of Brazil's electricity comes from low-carbon sources, such as hydropower, nuclear, solar, and wind power.
- Other countries with high percentages of low-carbon electricity include Paraguay, Iceland, Norway, and Nepal. These countries all have a significant amount of hydropower potential, which is a clean and renewable source of energy.
- Some countries, such as Bhutan and Costa Rica, have almost all of their electricity coming from low-carbon sources, even though they do not have as much hydropower potential as some of the other countries on the list. This is because they have been able to successfully invest in other renewable energy sources, such as solar and wind power.
- The graph also shows that many countries are still heavily reliant on fossil fuels for their electricity generation. For example, the United States gets only about 20% of its electricity from low-carbon sources. This means that the United States has a lot of work to do if it wants to reduce its greenhouse gas emissions and combat climate change.

Figure 3.14 Renewable Energy Generating Capacity PerCapita

- The graph shows the renewable electricity-generating capacity per person for several countries, from 2000 to 2021.
- It shows that Panama has the highest renewable electricity-generating capacity per capita in the region, with over 22,000 kilowatt-hours per person in 2021.
- Other countries with high renewable electricity-generating capacity per capita include Costa Rica, Brazil, Uruguay, and Colombia.
- The graph also shows that the renewable electricity-generating capacity per capita has increased in most countries over the past 20 years. For example, Panama's capacity has increased from around 2,000 kilowatt-hours per person in 2000 to over 22,000 kilowatt-hours per person in 2021.

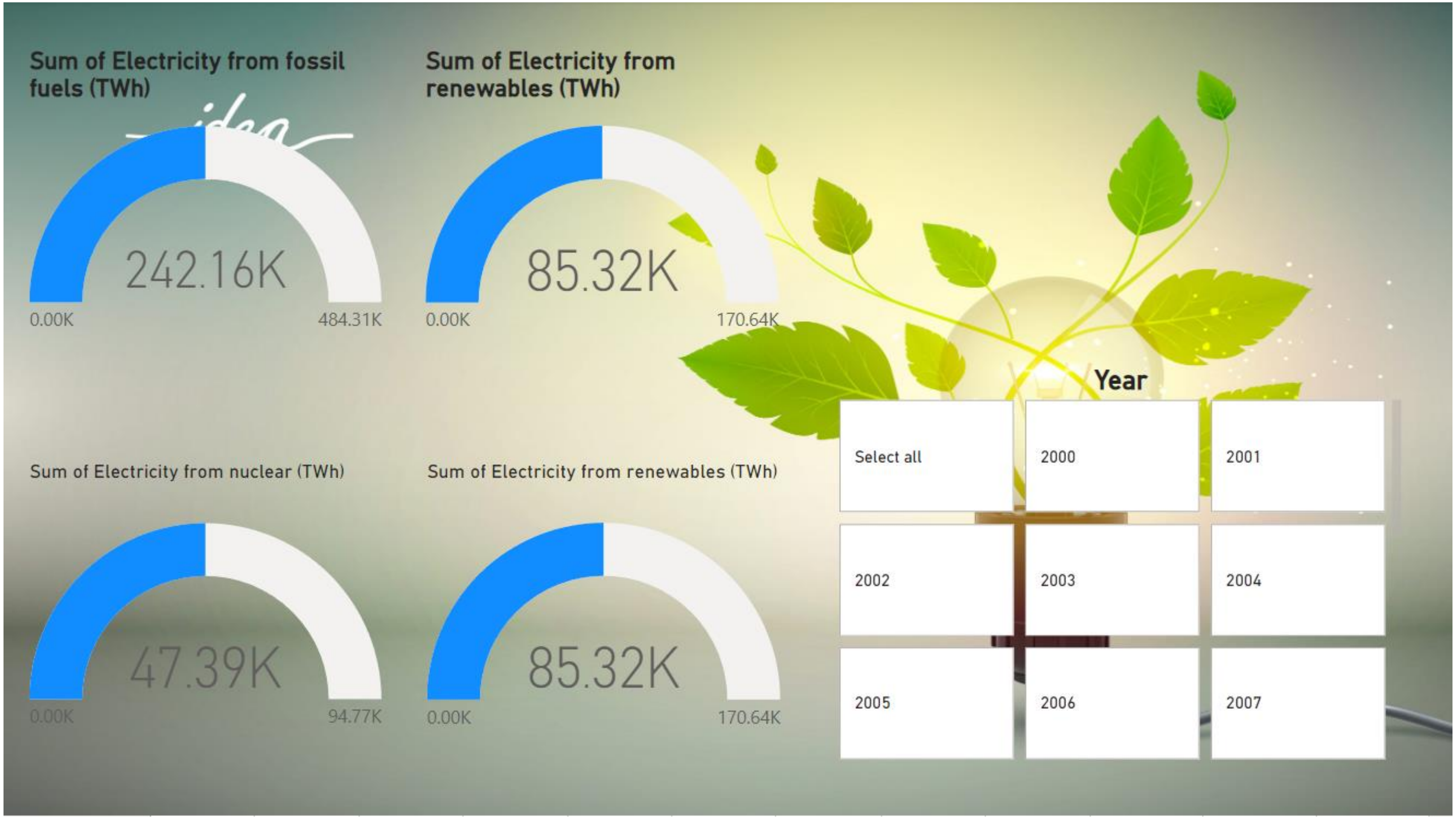


Figure 3.15 Renewable Energy Dashboard by Year Wise

From 2000 to 2020, the world's electricity landscape underwent a significant shift. Fossil fuels remained king, supplying a hefty 65% of our power in 2020. However, renewable energy sources like solar and wind were on the rise, steadily climbing from a mere 6% in 2000 to a respectable 20% by 2020. Meanwhile, nuclear power held steady at around 10%. While fossil fuels still reigned supreme, the rise of renewables paints a hopeful picture of a gradual transition towards a cleaner energy future

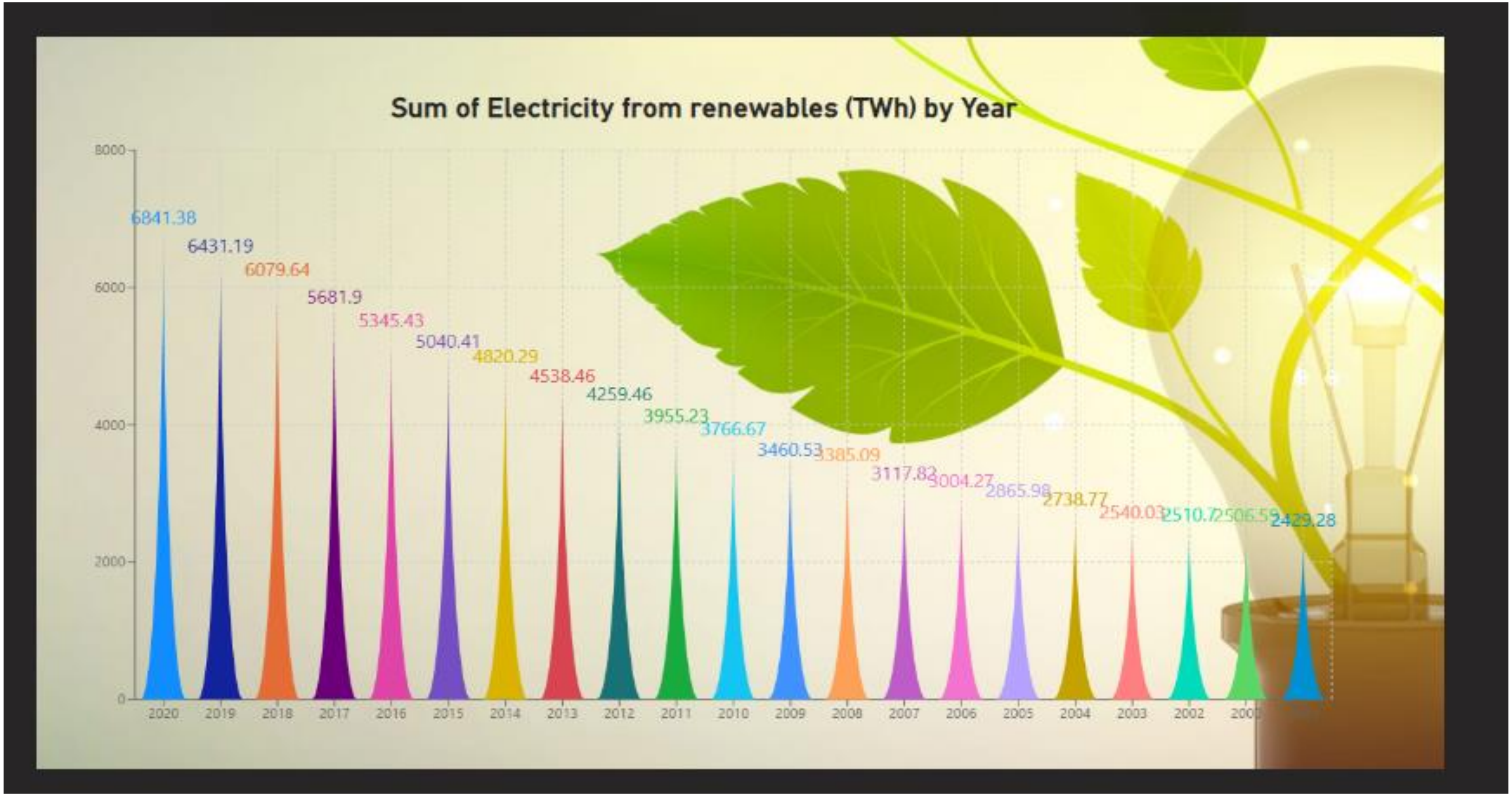


Figure 3.16 Sum of Electricity from Renewables (Twh) by year

Renewables are surging, From 2000 to 2020, the world witnessed a remarkable green surge in electricity generation. Renewables, starting from a humble 3,460 TWh in 2000, more than doubled to a mighty 8,004 TWh by 2020. The pace quickened in recent years, with impressive leaps compared to the early 2000s. Despite this phenomenal rise, renewables still make up only about 27% of the total electricity mix, with fossil fuels clinging to the top spot for now. This chart paints a hopeful picture of a world steadily tilting towards cleaner energy sources, but the race to dethrone fossil fuels is far from over.

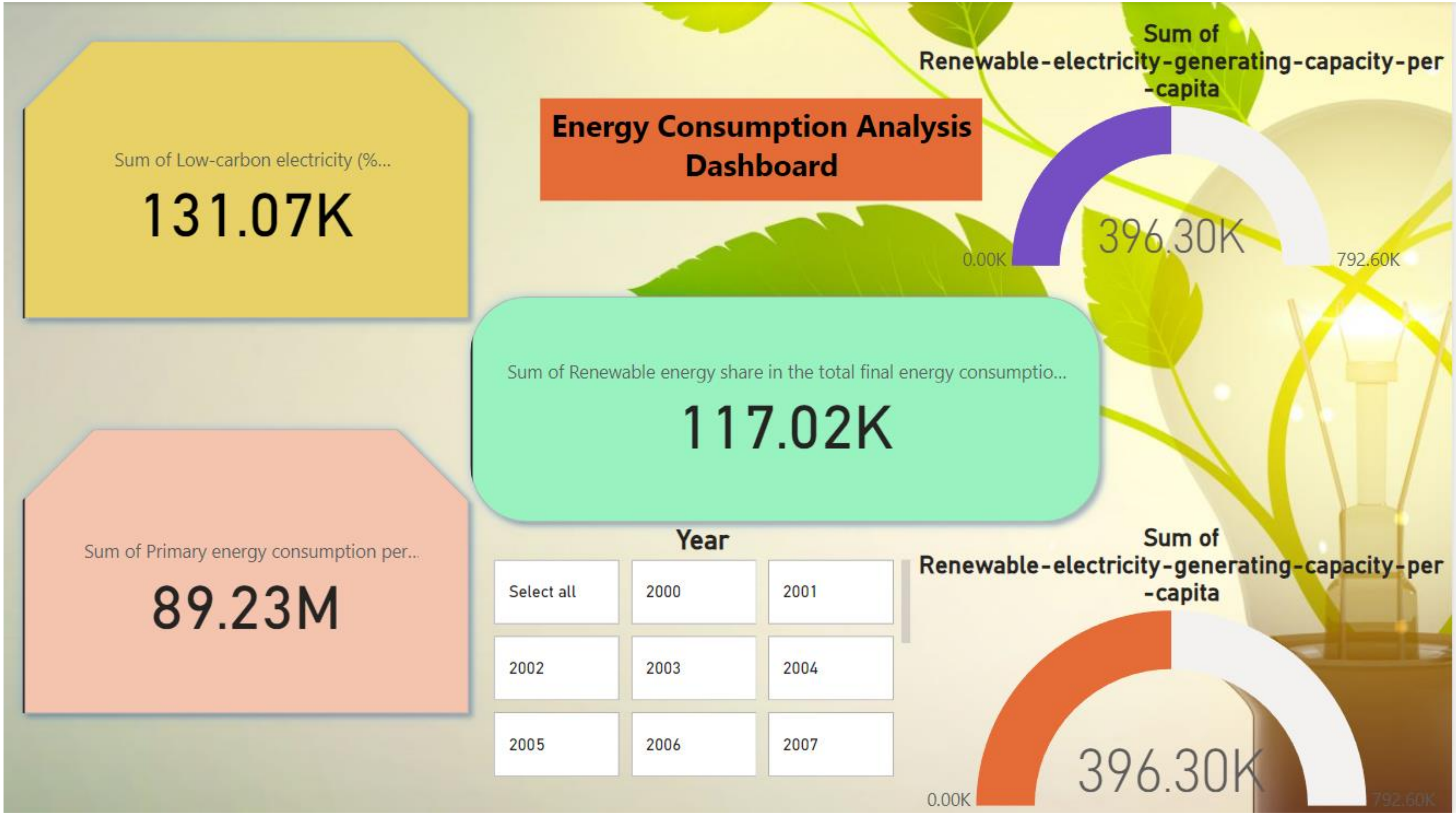


Figure 3.17 Energy Consumption Analysis Dashboard

- The graph shows that the sum of renewable energy values in the total solar energy consumption has increased steadily over time. In 2015, the sum of renewable energy values was around 70%, while in 2020, it had increased to over 117%.
- This means that renewables are playing an increasingly important role in meeting our solar energy needs. This is a positive trend, as it helps to reduce our reliance on fossil fuels and combat climate change.
- The graph also shows that there is a significant variation in the sum of renewable energy values from country to country. For example, in 2020, the sum of renewable energy values in Germany was over 400%, while in the United States it was only around 100%

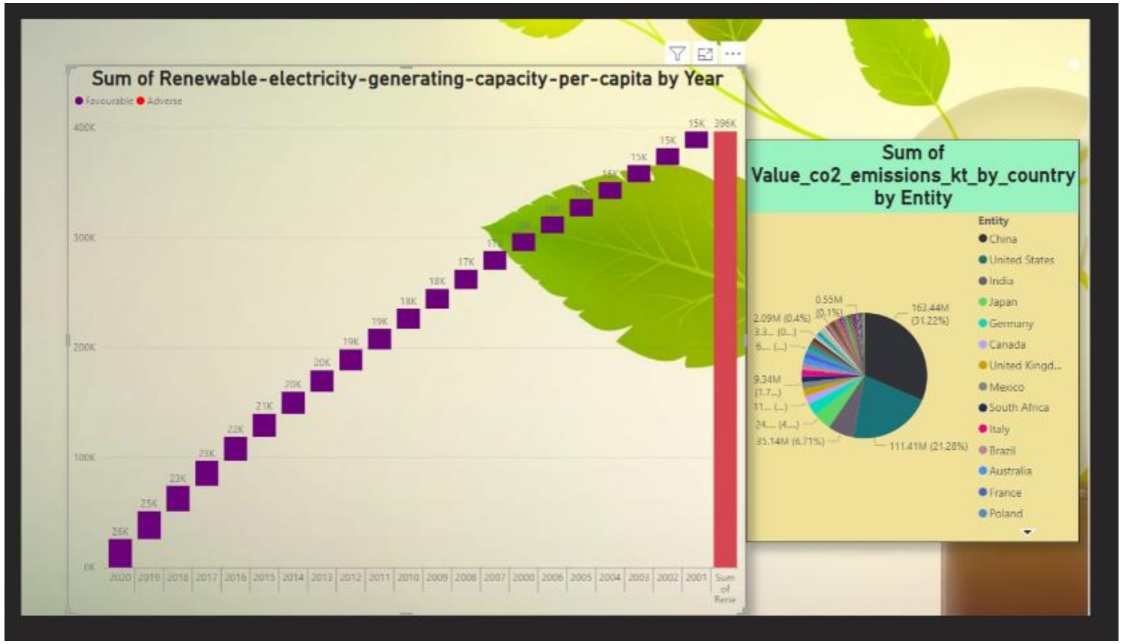


Figure 3.18 Year Wise Electricity Generation

- The world's energy landscape is undergoing a slow but steady transformation, as captured in this eye-opening graph. Fossil fuels, once the undisputed king of the energy hill, have seen their dominance wane from a peak of 87% in 1965 to a still-substantial but shrinking 81% by 2018. This gradual dethronement paves the way for a brighter future powered by cleaner sources.
- Meanwhile, renewable energy sources, once relegated to a niche corner, are basking in the spotlight of a steady ascent. From a humble 1% in 1965, their share has more than quadrupled to a respectable 14% by 2018. Solar panels sprouting on rooftops, wind turbines dotting landscapes, and geothermal plants tapping into the Earth's core heat – these clean energy champions are steadily claiming their rightful place in the global energy mix.
- Nuclear energy, the stoic veteran of the bunch, has maintained a steady presence around the 5% mark throughout this period. While its future trajectory remains a topic of debate, its current contribution to the energy mix cannot be overlooked.
- This shift in energy sources is not merely a statistical footnote; it holds profound implications for our planet's health. As we move away from fossil fuels and embrace cleaner alternatives, we take a critical step towards curbing climate change and building a more sustainable future. The road ahead may be long, but the progress we've witnessed so far is a testament to human ingenuity and a beacon of hope for a cleaner, greener tomorrow.
- So, let us continue to nurture the sun, harness the wind, and tap into the Earth's renewable bounty. The future of energy is bright, and with each watt generated from a clean source, we move closer to a world powered by the forces of nature, not the remnants of ancient carbon.

3.5 Dashboard creation

To commence the dashboard creation process for the sustainable energy analysis, users should access the Power BI tool from the right side of the Power BI home page. Upon logging in with their credentials, including ID and password, the next step entails publishing the document. This is achieved by selecting the 'Publish' button positioned on the right side of the window, as exemplified in the accompanying figure. This step marks the transition from data exploration to the visualization of sustainable energy trends and insights on a dynamic and interactive dashboard within the Power BI environment.

Then it displays the dialogue box like do you want to save changes in that click on the save button and it is shown in the figure below.

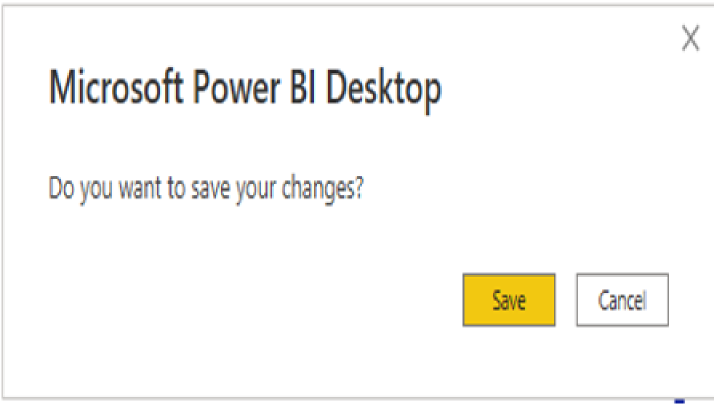


Figure 3.19 Saving the changes

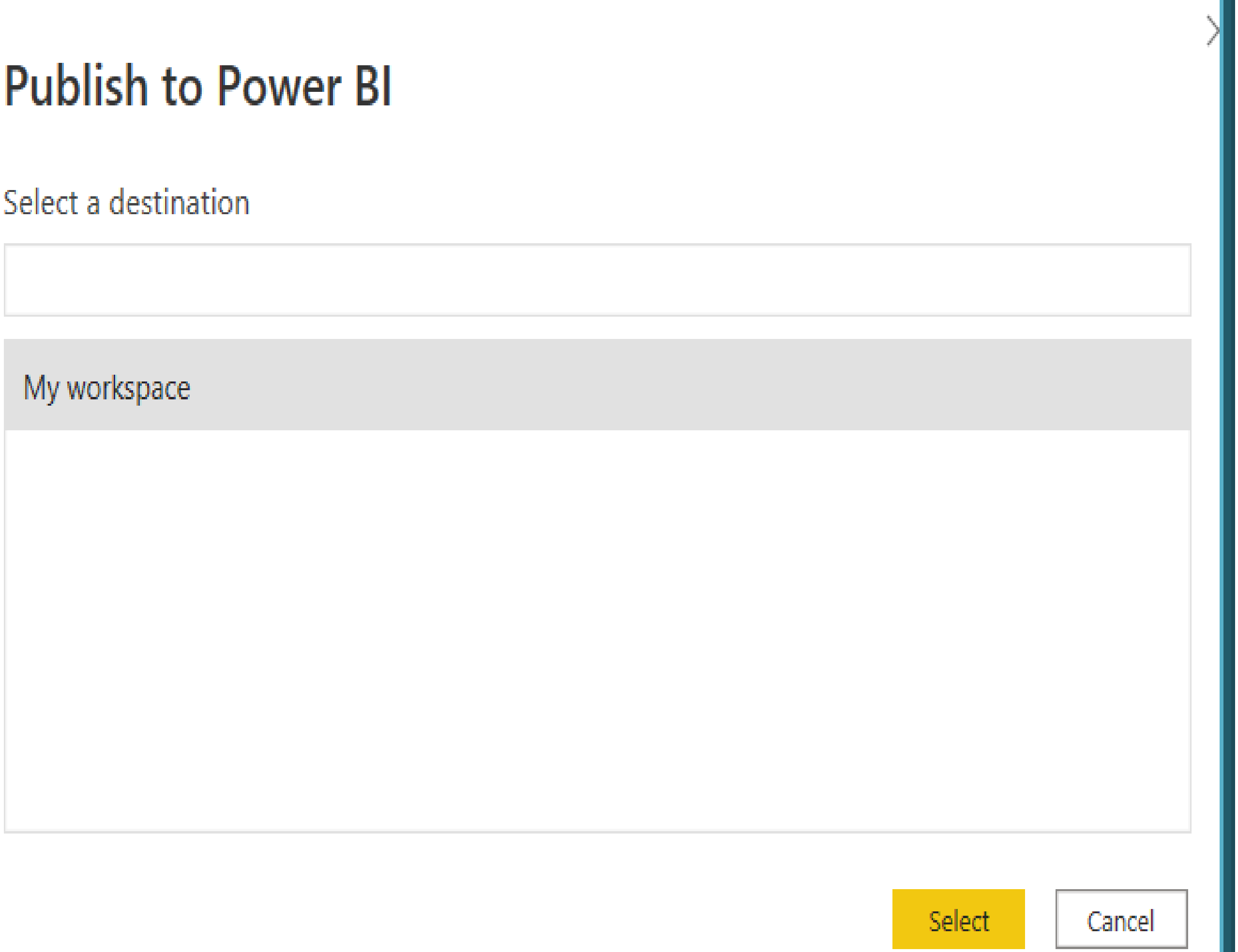


Figure 3.20 Selecting the workspace to publish

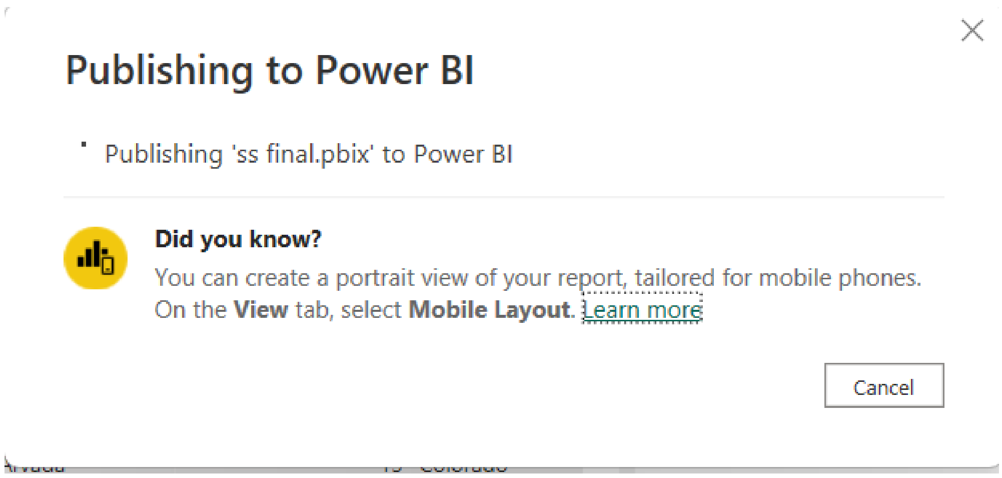


Figure 3.21 Publishing the queries, Dataset

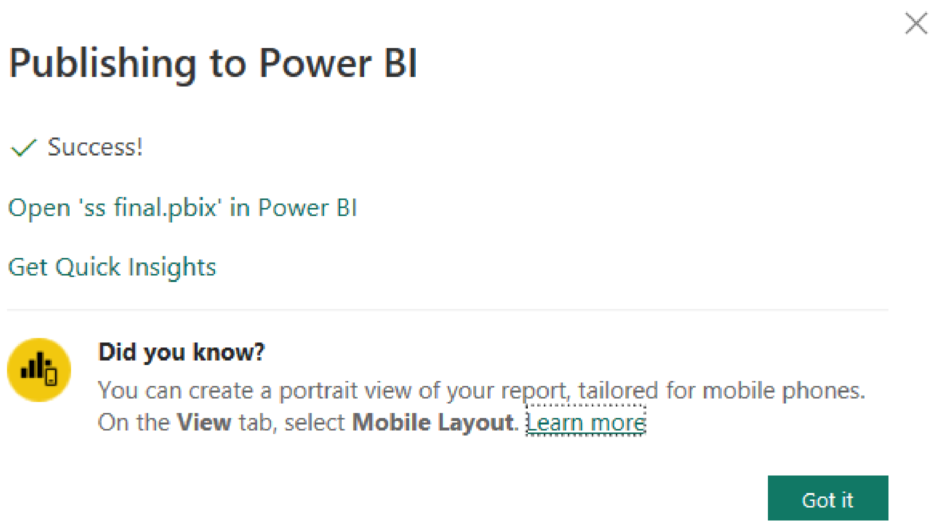


Figure 3.22 Opening in the Chrome

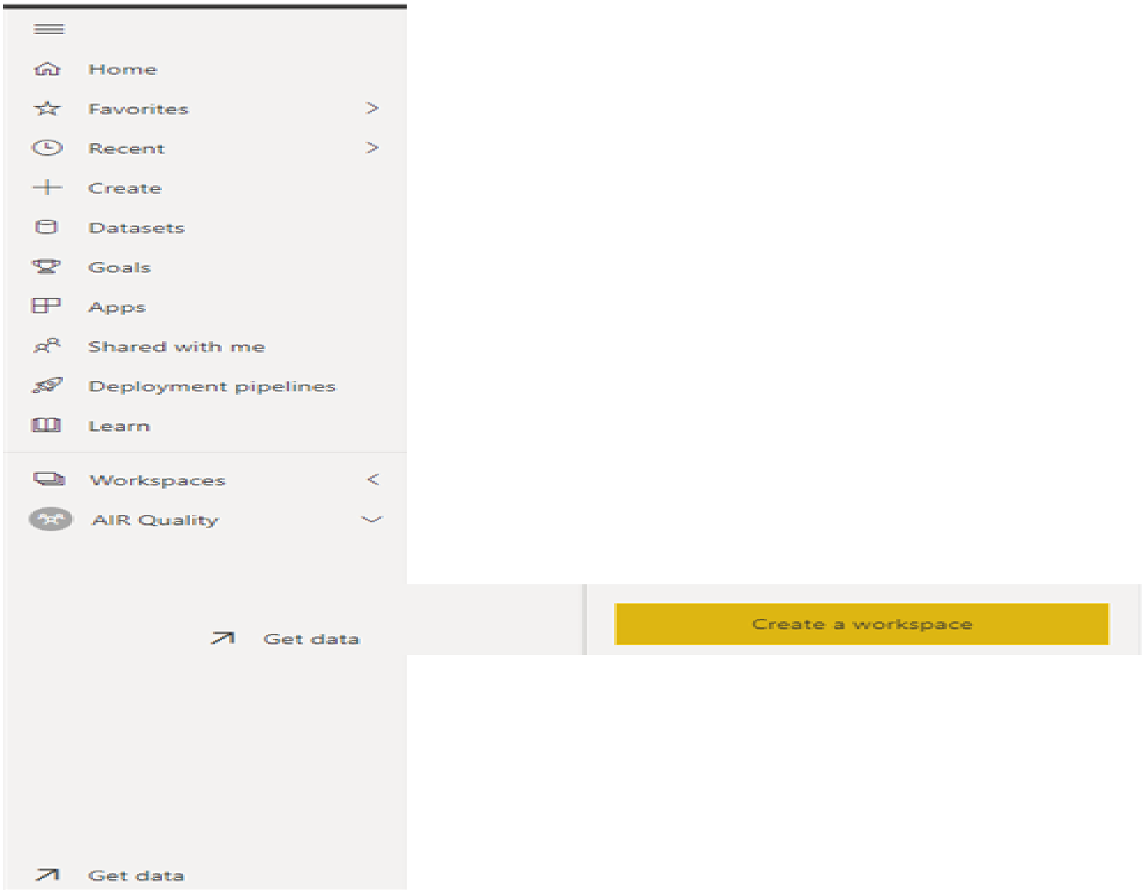


Figure 3.23 Creating the Workspace

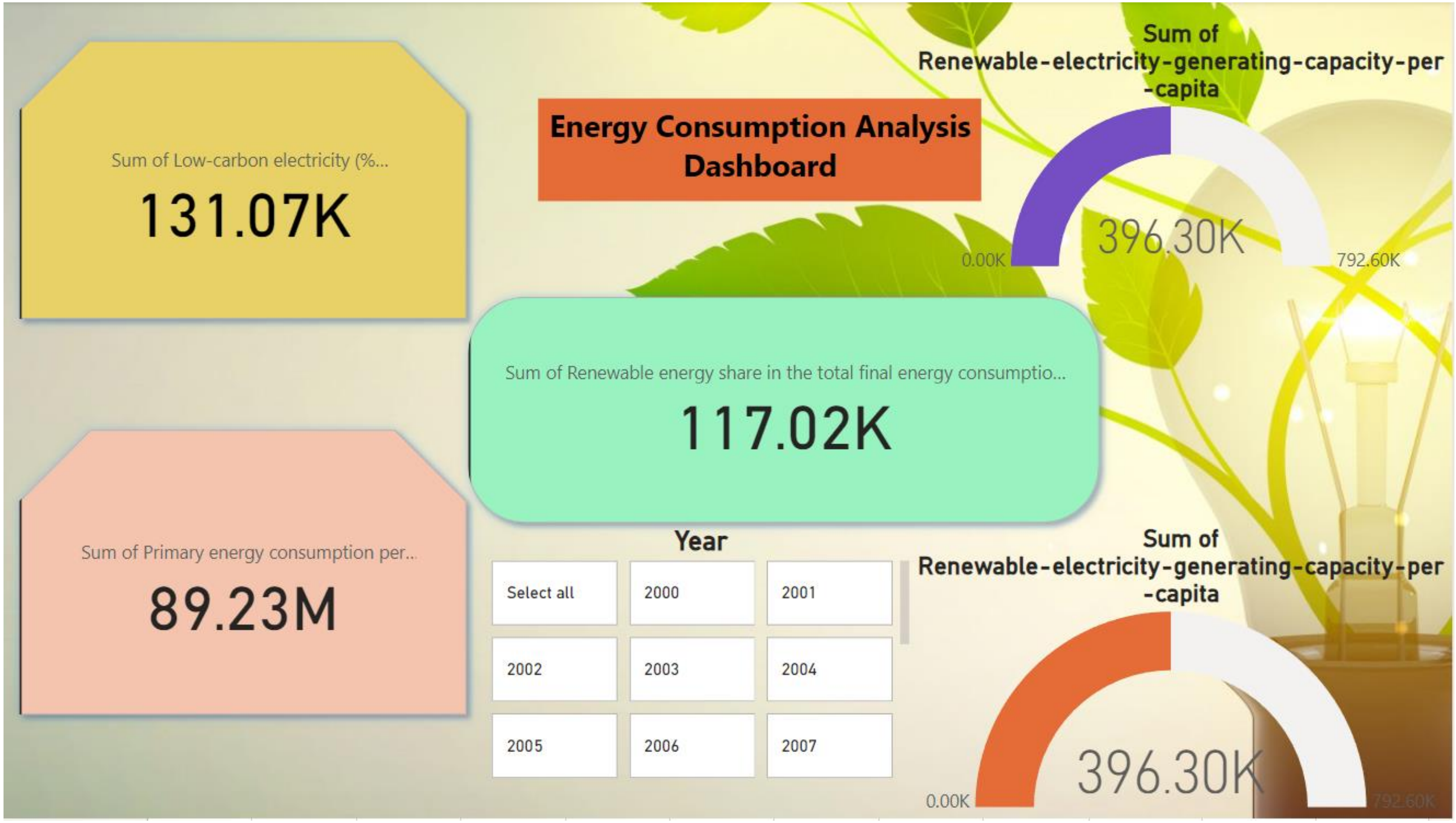


Figure 3.24 Project Dashboard 1

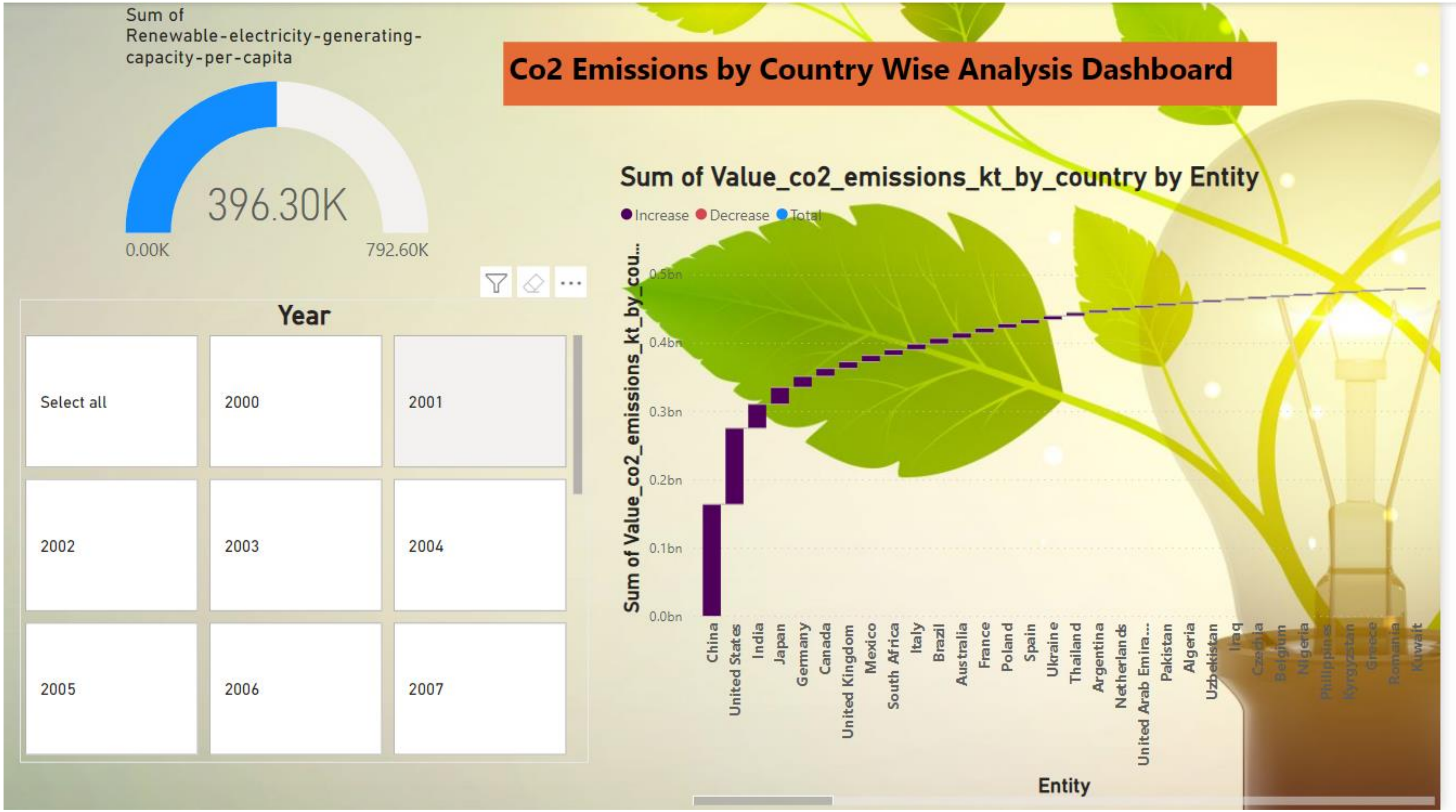


Figure 3.25 Project Dashboard 2

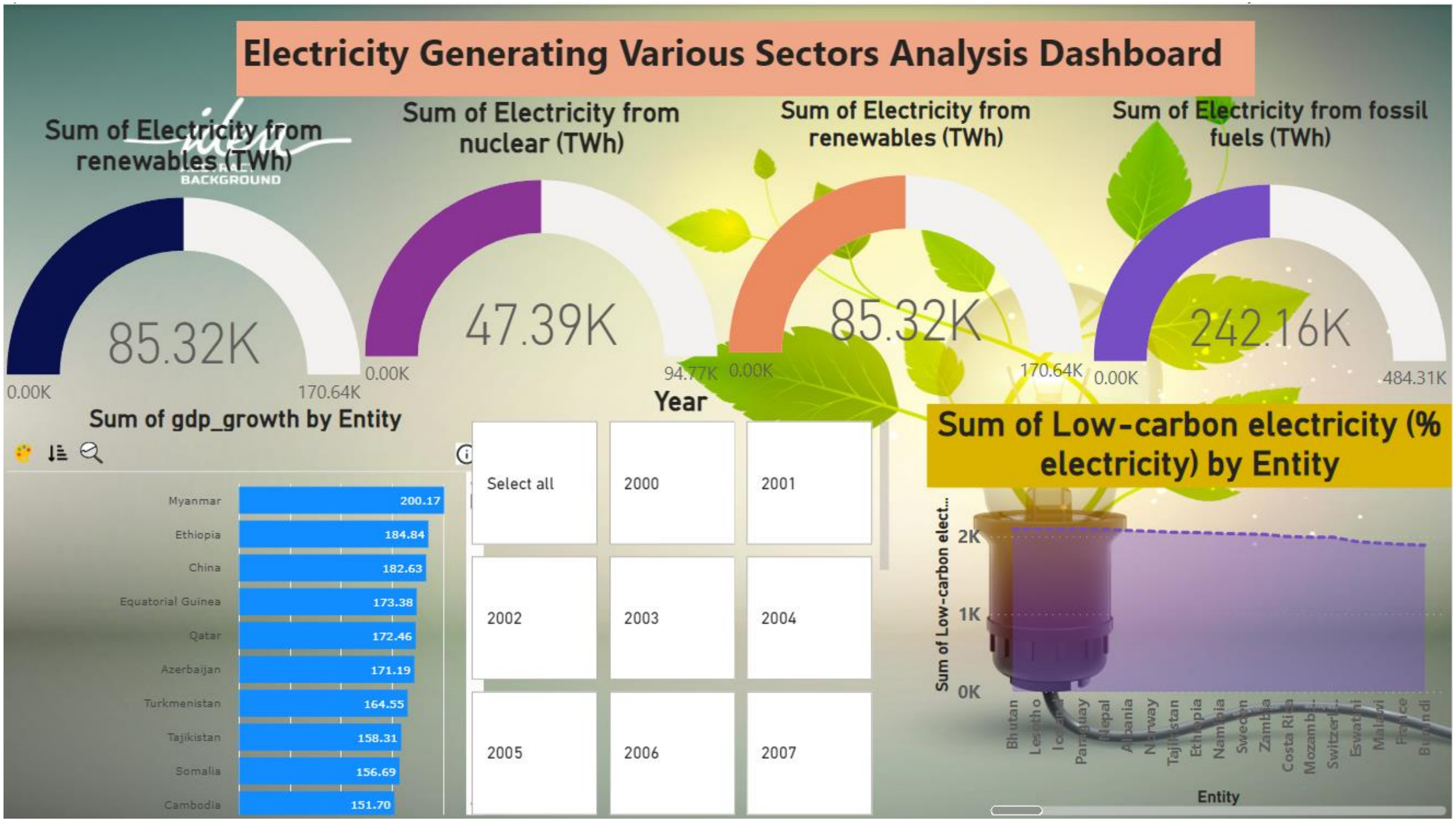


Figure 3.26 Project Dashboard 3

4 Reflective report

Acknowledging Kaggle, an open-source platform, and Microsoft Power BI for facilitating access to this significant sustainable energy dataset and analysis tools, the author underscores the dataset's importance for informing decisions in future energy initiatives. The analysis is instrumental in shaping sustainable energy policies and practices globally. The author employed diverse visual tools, including bar charts, pie charts, line charts, maps, and tree charts, to effectively convey insights and trends gleaned from the dataset, enhancing the understanding of sustainable energy dynamics and fostering informed decision-making for a more sustainable future.

5 Conclusions

The conclusion of the Power BI report on the sustainable energy dataset illuminates key insights derived from the analysis. Encompassing areas such as renewable energy distribution, demographic factors, historical trends, and global impact, the report underscores significant patterns and correlations. It offers a comprehensive understanding of sustainable energy trends and practices. Recommendations for further analysis are provided to delve into identified areas in greater detail. Overall, the Power BI report stands as a valuable tool for unraveling the intricacies of the sustainable energy dataset, providing insights that contribute to a deeper understanding of global sustainability efforts and opportunities for ongoing exploration.

6 References

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Kazmi, H., Munné-Collado, Í., Mehmood, F., Syed, T. A., & Driesen, J. (2021). Towards data-driven energy communities: A review of open-source datasets, models and tools. *Renewable and Sustainable Energy Reviews*, *148*, 111290.